流體輸送
(管線、控制閥、壓縮機及泵Head、NPSHa計算)

授課時數：3hr

講　　師：劉芳榮 (Dean Liu)
電子信箱：flliu@dcc.com.tw
連絡電話：07-3514151 Ext.258

課程大綱

1. 前言
2. 概論
3. 管線壓損計算
   3.1 單相液體流 管線壓損
   3.2 氣相流 管線壓損
   3.3 兩相流 管線壓損
4. 泵浦的揚程(Head)與(NPSHa)計算
5. 控制閥壓差計算
6. 流量計壓損
7. 參考資料

1. 前言

1. 泵浦消耗全球約20%的電能，消耗工業生產廠約25%至50%的能源，若現場人員只是粗估揚程與流量，其為確保泵浦能力足夠往往會加大安全係數，造成選擇過大泵浦與過高揚程。使許多泵浦系統運轉在效率差而耗能的情況，所以正確的設計也是一種節能減碳。

2. 在一般化工廠的建廠費用中，管線系統費用約佔全部的10~30%（有的書提供的數據為25~40%），若能選擇正確的尺寸不僅能確保操作的可靠更能節省管線費用(尤其特殊材質管線)。

1. 前言

3. 通常下列的情況值得做詳細的經濟分析
   - 長途管線(有例子)
   - 冷卻水主管線
   - 合金鋼管
   - 特殊管線，例如內襯耐火材料的管線

4. 對於短距離的管線，在選擇管徑上通常並沒有明顯的最佳值，一般典型的做法是以「合理的壓降和流速」來決定管線的尺寸，所得的結果大致上是令人滿意的。

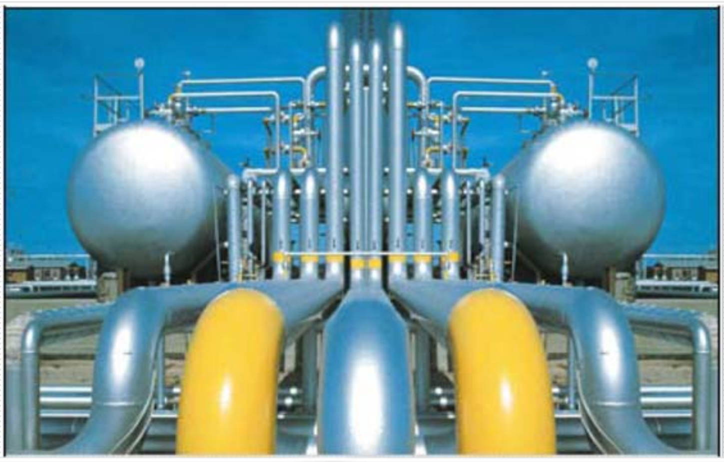
5. 管線的尺寸決定於管線的長度和可容許的壓降。

2. 概論

課程目的：

1. 選擇合適的管線尺寸與檢視特定的水力限制
2. 對於液體/氣體/兩相流體的管線壓損計算
3. 正確的應用設計基準與管線使用形式之方針
4. 範例練習，設計不是只憑感覺，目的讓課程參加者能有遵循的計算參考、方式或準則，得到所需的(管線、控制閥)壓損或泵浦揚程等

3.管線壓損計算



Picture from: PipeFlow.co.uk

3. 管線壓損計算

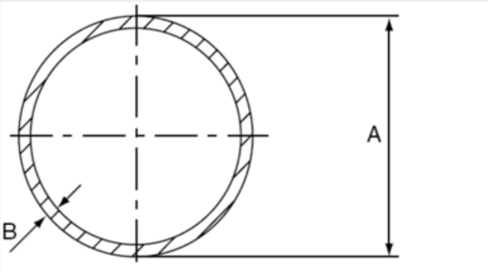
計算(Sizing)管線壓損所需的基本資料:

1.管材系統(Piping Code):

1.1.知道碳鋼(CS)或不銹鋼(SS)材質

1.2.)知道管壁厚度 (Wall Thickness)，才能求得管線的內徑(ID)。

Wall thickness and weights of piping manufactured to ASME B36.19



Taking the Dimensions of Steel Pipe

3. 管線壓損計算

管線的基本觀念:

1. 公稱尺寸（Nominal Pipe Size, NPS）是對管子尺寸的稱呼，為實際管徑之近似值

2.對於14”(含)以上 NPS是等於管線的外徑

NPS≧14”, NPS＝OD

3.考慮材料取得的問題，1-1/4”, 3-1/2”及5”儘可能不選用

4. Steel Pipe (Seamless and Welded):

Wall Thickness (WT) – STD, XS, XXS and per Schedules

5.Stainless Steel Pipe (Seamless and Welded):

Wall Thickness (WT) – 5S, 10S, 40S and 80S

3. 管線壓損計算

Dimensions of Welded and Seamless Pipe (Steel and SS)

		NOMINAL WALL THICKNESS													
NOMINAL PIPE	OUTSIDE DIAMETER	CARBON AND ALLOY STEELS ANSI STANDARD B36.10									STAINLESS STEELS ANSI STANDARD B36.19				NOMINAL PIPE
SIZE (in.)	(mm)	SCH. 10	SCH. 20	SCH. 30	STD	SCH. 40	EXTRA STRG.	SCH. 80	SCH. 160	XX STRG.	SCH. 5S	SCH. 10S	SCH. 40S	SCH. 80S	SIZE (in.)
1/2	21.34				2.77	2.77	3.73	3.73	4.78	7.47	1.65	2.11	2.77	3.73	1/2
3/4	26.67				2.87	2.87	3.91	3.91	5.56	7.82	1.65	2.11	2.87	3.91	3/4
1	33.40				3.38	3.38	4.55	4.55	6.35	9.09	1.65	2.77	3.38	4.55	1
1-1/2	48.26				3.68	3.68	5.08	5.08	7.14	10.16	1.65	2.77	3.68	5.08	1-1/2
2	60.33				3.91	3.91	5.54	5.54	8.74	11.07	1.65	2.77	3.91	5.54	2
3	88.90				5.49	5.49	7.62	7.62	11.13	15.24	2.11	3.05	5.49	7.62	3
4	114.3				6.02	6.02	8.56	8.56	13.49	17.12	2.11	3.40	6.02	8.56	4
6	168.3				7.11	7.11	10.97	10.97	18.26	21.95	2.77	3.76	7.11	10.97	6
8	219.1		6.35	7.04	8.18	8.18	12.70	12.70	23.01	22.23	2.77	3.76	8.18	12.70	8
10	273.1		6.35	7.80	9.27	9.27	12.70	15.09	28.58	25.40	3.40	4.19	9.27	12.70	10
12	323.9		6.35	8.38	9.53	10.31	12.70	17.48	33.32	25.40	3.96	4.57	9.53	12.70	12
14	355.6	6.35	7.92	9.53	9.53	11.13	12.70	19.05	35.71		3.96	4.78			14
16	406.4	6.35	7.92	9.53	9.53	12.70	12.70	21.44	40.49		4.19	4.78			16
18	457.2	6.35	7.92	11.13	9.53	14.27	12.70	23.83	45.24		4.19	4.78			18
20	508.0	6.35	9.53	12.70	9.53	15.09	12.70	26.19	50.01		4.78	5.54			20
22	558.8	6.35	9.53	12.70	9.53		12.70	28.58	53.98		4.78	5.54			22
24	609.6	6.35	9.53	14.27	9.53	17.48	12.70	30.96	59.54		5.54	6.35			24
26	660.4	7.92	12.70	0.00	9.53		12.70								26
28	711.2	7.92	12.70	15.88	9.53		12.70								28
30	762.0	7.92	12.70	15.88	9.53		12.70				6.35	7.92			30
32	812.8	7.92	12.70	15.88	9.53	17.48	12.70								32
34	863.6	7.92	12.70	15.88	9.53	17.48	12.70								34
36	914.4	7.92	12.70	15.88	9.53	19.05	12.70								36

3. 管線壓損計算

計算(Sizing)管線壓損所需的基本資料:

2.流體的操作條件與物性:

2.1.設計的流量

2.2.液體:密度(Density)、黏度(Viscosity)

2.3.氣體:密度、黏度、分子量(Mw)、操作壓力、操作溫度、壓縮因子(Z)、比熱比(Cp/Cv)

2.4.兩相流:液體與氣體相各別的密度、黏度、液體的表面張力(Surface tension)

3. 管線壓損計算

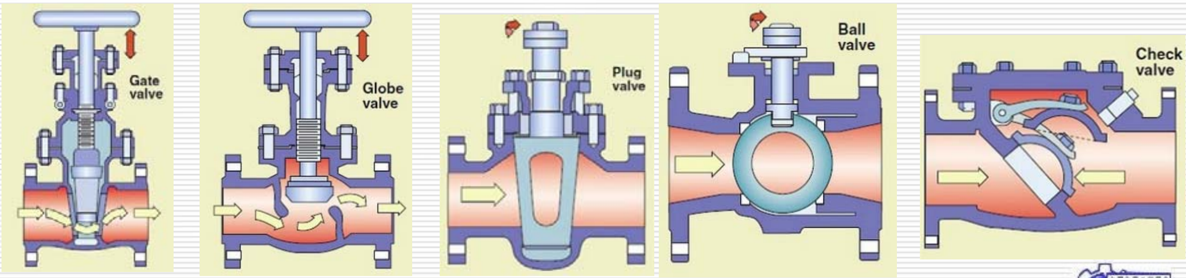
計算(Sizing)管線壓損所需的基本資料:

3.了解管線系統:

3.1.預估管線(Pipe)可能的長度(可從Plot Plan估算)、管線上可能管件(Fitting)的數量(90° Elbow/ 45° Elbow/ Tee/ Reducer等)。

3.2.管線上閥的種類(Gate/ Globe/ Plug/ Ball/ Check Valve)與數量(可從P &ID算出)。

3.3.若是Rating既有工廠則從實際的配管圖(ISO)逐一統計。



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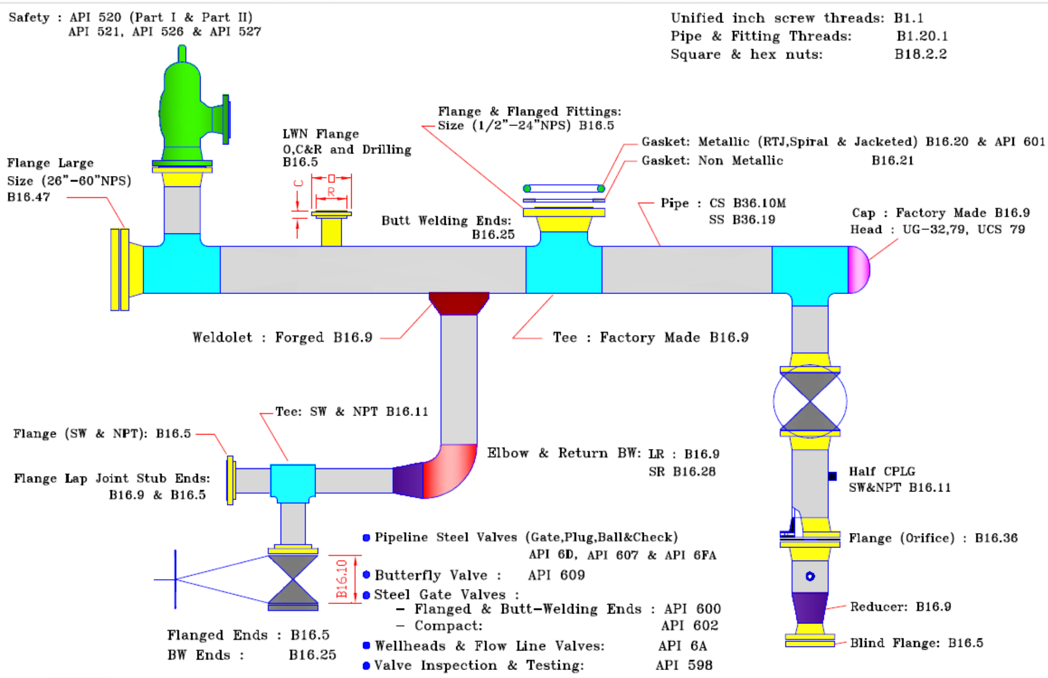
3. 管線壓損計算

計算(Sizing)管線壓損所需的基本資料:

3.4.了解管線系統:



3. 管線壓損計算

計算(Sizing)管線壓損所需的基本資料:

3.5. 管線的粗糙度(Roughness) 。

Roughness (mm)		
Glass, Cu, Al, Plastic,..	*New*	*0--0.0015*
Drawn steel,	*All case*	*0.001--0.05*
Welded steel,	*New*	*0.05--0.1*
	Slight rust	*0.15--0.2*
	Heavy rust	*up to 3*
Galvanized iron,	*New*	*0.12--0.15*

3. 管線壓損計算

計算(Sizing)管線壓損的方法:

1.查表法:圖表僅針對Water，簡單，但無對密度與黏度作補償亦無考慮Fitting，僅供參考。

FLOWSERVE Cameron Hydraulic Data

Friction of Water — **Asphalt-dipped Cast Iron and New Steel Pipe** (Based on Darcy's Formula) *(Continued)*

3 Inch

Flow U S gal per min	Asphalt-dipped cast iron, 3.0" inside dia: Velocity ft per sec	Velocity head ft	Head loss ft per 100 ft	Std wt steel sch 40, 3.068" inside dia: Velocity ft per sec	Velocity head ft	Head loss ft per 100 ft	Extra strong steel sch 80, 2.900" inside dia: Velocity ft per sec	Velocity head ft	Head loss ft per 100 ft	Schedule 160—steel, 2.624" inside dia: Velocity ft per sec	Velocity head ft	Head loss ft per 100 ft
10	.454	.00	**.042**	.434	.003	**.038**	.49	.00	**.050**	.593	.005	**.080**
15	.681	.01	**.088**	.651	.007	**.077**	.73	.01	**.101**	.890	.012	**.164**
20	.908	.01	**.149**	.868	.012	**.129**	.97	.02	**.169**	1.19	.022	**.275**
25	1.13	.02	**.225**	1.09	.018	**.192**	1.21	.02	**.253**	1.48	.034	**.411**
30	1.36	.03	**.316**	1.30	.026	**.267**	1.45	.03	**.351**	1.78	.049	**.572**
35	1.59	.04	**.421**	1.52	.036	**.353**	1.70	.04	**.464**	2.08	.067	**.757**
40	1.82	.05	**.541**	1.74	.047	**.449**	1.94	.06	**.592**	2.37	.087	**.933**
45	2.04	.06	**.676**	1.95	.059	**.557**	2.18	.07	**.734**	2.67	.111	**1.16**
50	2.27	.08	**.825**	2.17	.073	**.676**	2.43	.09	**.860**	2.97	.137	**1.41**
55	2.50	.10	**.990**	2.39	.089	**.776**	2.67	.11	**1.03**	3.26	.165	**1.69**
60	2.72	.12	**1.17**	2.60	.105	**.912**	2.91	.13	**1.21**	3.56	.197	**1.99**
65	2.95	.14	**1.36**	2.82	.124	**1.06**	3.16	.15	**1.40**	3.86	.231	**2.31**
70	3.18	.16	**1.57**	3.04	.143	**1.22**	3.40	.18	**1.61**	4.15	.268	**2.65**
75	3.40	.18	**1.79**	3.25	.165	**1.38**	3.64	.21	**1.83**	4.45	.307	**3.02**
80	3.63	.21	**2.03**	3.47	.187	**1.56**	3.88	.23	**2.07**	4.75	.350	**3.41**
85	3.86	.23	**2.28**	3.69	.211	**1.75**	4.12	.26	**2.31**	5.04	.395	**3.83**
90	4.08	.26	**2.55**	3.91	.237	**1.95**	4.37	.29	**2.58**	5.34	.443	**4.27**
95	4.31	.29	**2.83**	4.12	.264	**2.16**	4.61	.33	**2.86**	5.63	.493	**4.73**
100	4.54	.32	**3.12**	4.34	.293	**2.37**	4.85	.36	**3.15**	5.93	.546	**5.21**
110	4.99	.39	**3.75**	4.77	.354	**2.84**	5.33	.44	**3.77**	6.53	.661	**6.25**

From Flowserve Cameron Hydraulic Data

Friction Loss for Water – Sched 40 Steel Pipe

U.S. Gallons per Minute	3 In. (3.068" I.D.): V	$\frac{V^2}{2g}$	h_f
30			
35			
40			
50	2.17	0.073	0.66
60	2.60	0.105	0.92
80	3.47	0.187	1.57
100	4.34	0.293	2.39
120	5.21	0.421	3.37
140	6.08	0.574	4.51
160	6.94	0.749	5.81
180	7.81	0.948	7.28
200	8.68	1.17	8.90
220	9.55	1.42	10.7
240	10.4	1.69	12.6
260	11.3	1.98	14.7
280	12.2	2.29	16.9
300	13.0	2.63	19.2
350	15.2	3.57	26.3
400	17.4	4.68	33.9
500	21.7	7.32	52.5
600	26.0	10.5	74.8
700	30.4	14.3	101
800	34.7	18.7	131
1000			

From ITT Technical Data

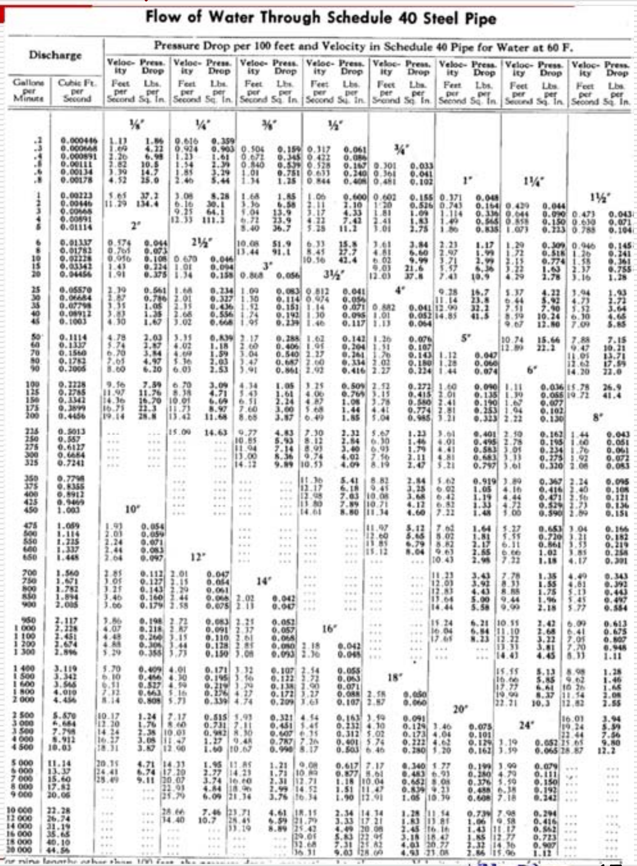
3. 管線壓損計算方法

計算(Sizing)管線壓損的方法:

2.查表法:

Flow of Water Through Schedule 40 Steel Pipe

Pressure Drop per 100 ft and Velocity in Schedule 40 Pipe for Water at 60F


Flow of Water Through Schedule 40 Steel Pipe

Pressure Drop per 100 feet and Velocity in Schedule 40 Pipe for Water at 60 F.

Discharge: Gallons per Minute; Cubic Ft. per Second — Velocity Feet per Second; Press. Drop Lbs. per Sq. In.

Flow of Fluids through Valves, Fittings, and Pipe, 1988, p.B-14

3. 管線壓損計算方法

管線(液體/氣體/兩相流)壓損主要是分兩部份:

1.計算管線的壓損:

1.1.計算雷諾數以確認流態(Laminar或Turbulent)

1.2.查圖磨擦係數(Friction Factor)

1.3.求單位管線長度的壓損值

1.4.確認實際的管線長度並算出壓損

2.由Fitting(管件型式與閥種類)數量計算:

2.1.計算出 K Factor of Fitting

2.2.計算出Fitting相當管長(Equiv. Length of fittings)

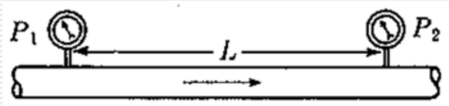
管線壓損 = 1. + 2.

3.1 液體管線壓損計算

液體管線(重力流/泵浦輸送)壓損計算:

1.計算管線的壓損:

1.1. Darcy's formula



$$h_L = f\frac{L}{D}\frac{v^2}{2g} \qquad v = \frac{q}{A} = \frac{w}{A\rho} = \frac{w\bar{V}}{A}$$

h_L :壓損(m-Liqiud), g : 重力加速度 (9.8 m/s/s)

L :管線長度(m) , w : 流量 (kg/hr)

D :管線內徑(mm), A :管線截面積

v :管線內流體的平均流速(m/s) , ρ : 重量密度 (kg/m³)

f :磨擦係數(Friction factor)

3.1 液體管線壓損計算

2.1.計算雷諾數以確認流態(Laminar或Turbulent)

2.1.1 Reynolds Number, Re

雷諾數是一個沒有單位的數值(無因次),是用來表示慣性力與黏度的比值

$$R_e = \frac{Dv\rho}{\mu_e} = \frac{Dv\rho}{32.2\mu'_e} = 123.9\frac{dv\rho}{\mu}$$

v:平均流速 (m/s)

D:管線直徑 (mm)

μ_e:流體黏度(cp)

ρ:流體密度(kg/m³)

3.1 液體管線壓損計算

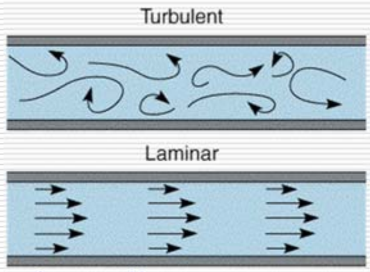
2.1計算雷諾數以確認流態(Laminar或Turbulent)

2.1.2 Reynolds Number, Re

$$R_e = \frac{Dv\rho}{\mu_e}$$

當Re<2100

為Laminar層流(又可稱作黏滯流動、線流)狀態



當2100<Re<4000

為過渡流狀態(Critical Zone)

當Re>4000

為Turbulent湍流(又可稱作紊流、擾流)狀態

3.1 液體管線壓損計算

2.1.3.確認流態後計算 f :磨擦係數(Friction factor)

當Re<2100為Laminar層流(一般很少出現層流狀態)

f :磨擦係數(Friction factor)

$$f = \frac{64}{R_e} = \frac{64\mu_e}{Dv\rho} = \frac{64\mu}{124dv\rho}$$

3.1 液體管線壓損計算

2.1.4.計算確認流態計算 f :磨擦係數(Friction factor)

當Re>4000為Turbulent湍流

f :磨擦係數 (Friction factor)包含湍流與過渡流狀態

此時磨擦係數就不僅只是簡單的雷諾數關係式也需考慮相對粗糙度(Relative Roughness, ε/D)

ε:管壁粗糙度

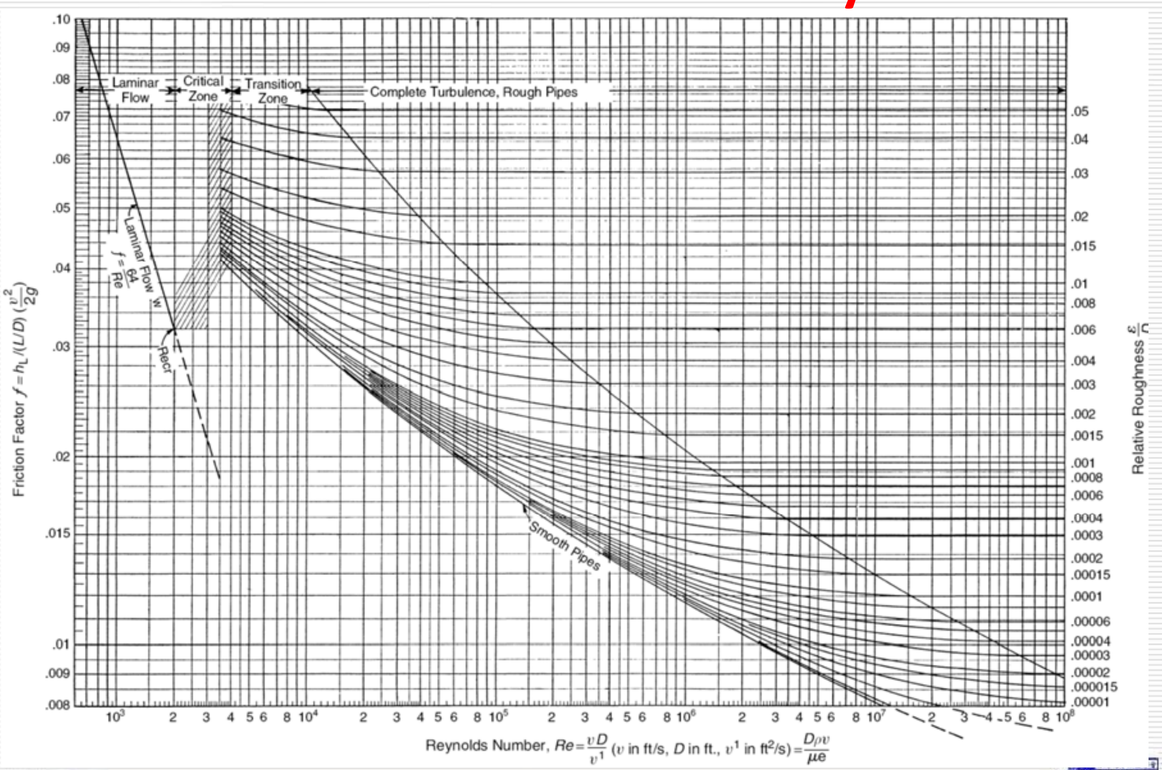
D:管線直徑

查圖(如下頁)得磨擦係數：

L. F. Moody Chart (從Darcy Formula整理出並被廣泛的使用於Friction Factor)

3.1 液體管線壓損計算

查圖:求得friction Factor Moody



3.1 液體管線壓損計算(長春)

2.2計算確認流態計算 f :磨擦係數(Friction factor)

當Re > 4000為Turbulent湍流

f :磨擦係數 (Friction factor)

磨擦係數:不僅使是簡單的雷諾數關係式也要考慮相對粗糙度(Relative Roughness, ε/D)

$$f = (A - \frac{(B-A)^2}{C-2B+A})^{-2} \qquad A = -2.0x\log(\frac{\varepsilon/D}{3.7} + \frac{12}{\mathrm{Re}})$$

$$B = -2.0x\log(\frac{\varepsilon/D}{3.7} + \frac{2.51A}{\mathrm{Re}}) \qquad C = -2.0x\log(\frac{\varepsilon/D}{3.7} + \frac{2.51B}{\mathrm{Re}})$$

Chem. Eng., March 5, 1984., p.63~64 Estimate friction factor accurately

3.1 液體管線壓損計算(長春)

2.3液體管件(Valve and Fittings)壓損計算:

2.3計算管件的壓損(同管線公式):

2.3.1. Darcy's formula

$$h_L = (f\frac{L}{D})\frac{v^2}{2g} \qquad h_L = K\frac{v^2}{2g}$$

h_L :壓損(m-Liqiud), g : 重力加速度 (9.8 m/s/s)

L :管線長度(m) , D :管線內徑(mm),

L:KD/f 相當管長(Equivalent Length)

v :管線內流體的平均流速(m/s) ,

K :抗阻係數(Resistance Coefficient)，含Valve and Fittings

3.1 液體管線壓損計算(長春)

2.3.2. Two-K Mothod計算Fitting K factor(考慮Re與管徑)

$K=K_1/Re + K_\infty (1+25.4/ID)$

Chem. Eng. Aug. 24, 1981, p.96~100
The two-K method predicts head losses in pipe fittings

其中: K_1=K for the fitting at Re=1

K_∞ =K for a large fitting at Re=∞

ID=ID of pipe (mm)

- ✓ 當Re足夠大時，K值不受Re影響
- ✓ 當Re降至1000時，K值開始上升
- ✓ 當Re降至100時， K值與Re呈反比
- ✓ 事實上小尺寸管件對表面粗糙度與突然改變截面積是較敏感的，往往對於小尺寸管件K值較高。

可以精確地應用於大管徑 & Alloy Fitting及低Reynolds常數

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3.1 液體管線壓損計算(長春)

2.3.2. Two-K Mothod計算Fitting K factor

$K=K_1/Re + K_\infty (1+25.4/ID)$

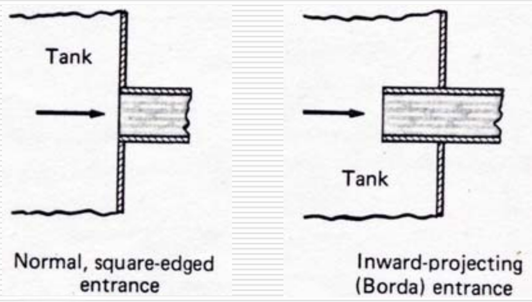
其中: K_1=K for the fitting at Re=1

K_∞ =K for a large fitting at Re=∞

ID=ID of pipe (mm)

2.3.3. Two-K Mothod計算(Pipe Entrance and Exit) K factor

$K=K_1/Re + K_\infty$



可以精確地應用於大管徑 & Alloy Fitting及低Reynolds常數

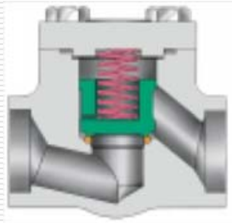
3.1 液體管線壓損計算(長春)

2.3.3. Constants for two-K Method

Chem. Eng. Aug. 24, 1981, p.96~100
The two-K method predicts head losses in pipe fittings

TWO K METHOD to predict head loss in fittings

	K1	K ∞		K1	K ∞
ELBOW			**TEE :**		
90°ANGLE:	K1	K∞	**Used as elbow:**		
standard, screw	800	0.4	standard, screw	500	0.7
standard, flange/weld	800	0.25	long-radius, screw	800	0.4
long-radius, all types	800	0.2	standard, flange/weld	800	0.8
metered elbows			stub-in type branch	1000	1
1 weld 90	1000	1.15	**Run through tee :**		
2 weld 45	800	0.35	screw	200	0.1
3 weld 30	800	0.3	flange/weld	150	0.5
4 weld 22-1/2	800	0.27	stub-in branch	100	0
5 weld 18	800	0.25			
45°ANGLE:			**VALVES :**		
standard, all types	500	0.2	**Gate, ball, plug :**		
long-radius, all types	500	0.15	full line size, B=1.0	300	0.1
metered, 1 weld 45	500	0.25	reduced trim, B=0.9	500	0.15
metered, 2 weld 22-1/2	500	0.15	reduced trim, B=0.8	1000	0.25
180°ANGLE:			Globe, standard	1500	4
standard, screw	1000	0.6	Globe, angle/ Y-type	1000	2
standard, flange/weld	1000	0.35	Diaphragm, dam type	1000	2
long-radius, all types	1000	0.3	Butterfly	800	0.25
PIPE ENTRANCE, normal	160	0.5	**Check, lift**	**2000**	**10**
PIPE ENTRANCE, Borda	160	1	swing	1500	1.5
PIPE EXIT	0	1	tilting-disk	1000	0.5



注意:
管徑 1-1/2" (含)以下, 使用 Lift type check valves, 務必要考慮, 因其壓降可能很大

3.1 液體管線壓損計算

範例:管線壓損計算

✓ **P-259泵浦入口:**

流量為20,000 kg/hr，密度(Density)為990 kg/m³，流體的黏度(Viscosity)為0.697 cp ，4" S40S管線，管線長度為20米長，有四個90° Elbow，一個Tee (Used as elbow)，1顆手動閥Gate Valve。

3.1 液體管線壓損計算

範例:管線與管件壓損計算

1.計算雷諾數(Re)以確認流態(Laminar或Turbulent)

$$R_e = \frac{Dv\rho}{\mu_e} = \frac{(102.26)(0.683)(990)}{(0.697)} = 99203$$

2. Re>4000判定為Turbulent湍流

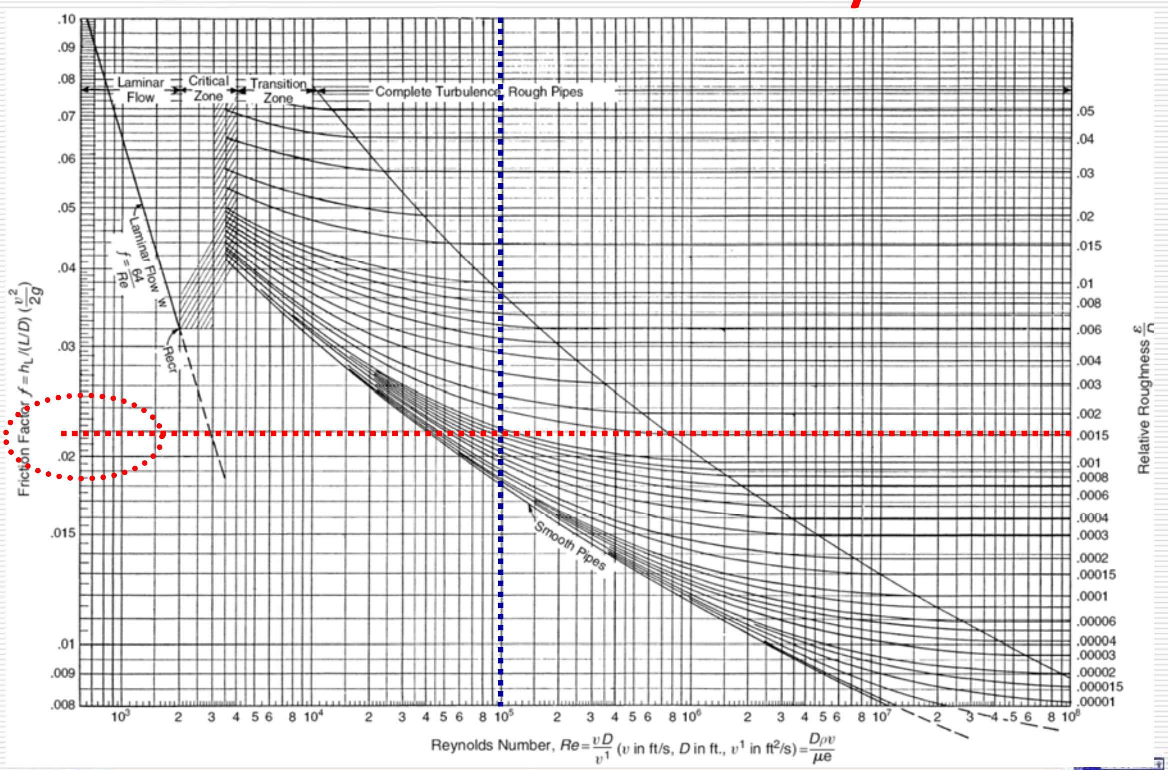
3.查圖求出管線之friction factor = 0.0195

4.相對粗糙度(Relative Roughness, ε/D)

ε/D=0.1/102.26=0.000978

3.1 液體管線壓損計算

查圖:求得friction factor　　Moody



3.1 液體管線壓損計算

範例:管線(Friction Factor查圖)與管件壓損計算

1.Darcy's formula:

$$h_{L(pipe)} = f\frac{L}{D}\frac{v^2}{2g} \qquad h_{L(Fitting\_and\_Valve)} = (f\frac{L}{D})\frac{v^2}{2g} = K\frac{v^2}{2g}$$

2.管線與管件壓損(m-Liquid):H_L= $H_{L(Pipe)}$ + $H_{L(Fitting + Valve)}$

$$h_L = (f\frac{L}{D}+K)\frac{v^2}{2g} = (0.0195x\frac{20}{(102.26/1000)}+3.667)\frac{(0.683)^2}{2x9.8}$$

→ **H_L= 0.1780 m-Liquid**

→ **DP_{piping} = 0.1780m x 990 kg/m³ / 10000(cm²/m²)**

= 0.0176 (kg/cm²)

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3.1 液體管線壓損計算(長春)

範例:管線與管件壓損計算

1.計算雷諾數(Re)以確認流態(Laminar或Turbulent)

$$R_e = \frac{Dv\rho}{\mu_e} = \frac{(102.26)(0.683)(990)}{(0.697)} = 99203$$

2. 求出管件(Fitting)之K(from Two-K Method)

ELBOW	K1_fit	K2_fit	Qty	K
90°ANGLE :	K_1	K_∞		
long-radius, all types	800	0.2	4	1.031
TEE :				
Used as elbow :				
standard, flange/weld	800	0.8	1	1.007
VALVES :				
Gate, ball, plug :				
full line size, B=1.0	300	0.1	1	0.128
PIPE ENTRANCE, normal	160	0.5	1	0.502
PIPE EXIT	0	1	1	1.000
**** SUM ****				3.667

3.1 液體管線壓損計算(長春)

範例:長春管線與管件壓損計算

1.計算雷諾數(Re)以確認流態(Laminar或Turbulent)

2. Re>4000判定為Turbulent湍流

3. 相對粗糙度(Relative Roughness, ε/D)

ε/D=0.1/102.26=0.000978

$$f=(A-\frac{(B-A)^2}{C-2B+A})^{-2}$$

$$A=-2.0x\log(\frac{\varepsilon/D}{3.7}+\frac{12}{\mathrm{Re}})$$

$$B=-2.0x\log(\frac{\varepsilon/D}{3.7}+\frac{2.51A}{\mathrm{Re}})$$

$$C=-2.0x\log(\frac{\varepsilon/D}{3.7}+\frac{2.51B}{\mathrm{Re}})$$

3.1 液體管線壓損計算(長春)

範例:長春管線與管件壓損計算

4.計算friction factor

$$A=-2.0x\log(\frac{\varepsilon/D}{3.7}+\frac{12}{\mathrm{Re}})=-2.0x\log(\frac{0.000978}{3.7}+\frac{12}{99203})=6.8284$$

$$B=-2.0x\log(\frac{\varepsilon/D}{3.7}+\frac{2.51A}{\mathrm{Re}})=-2.0x\log(\frac{0.000978}{3.7}+\frac{2.51x6.8284}{\mathrm{Re}})=6.7189$$

$$C=-2.0x\log(\frac{\varepsilon/D}{3.7}+\frac{2.51B}{\mathrm{Re}})=-2.0x\log(\frac{0.000978}{3.7}+\frac{2.51x6.7189}{\mathrm{Re}})=6.7244$$

$$f=(A-\frac{(B-A)^2}{C-2B+A})^{-2}=0.022117$$

求出管線之f = 0.0221

3.1 液體管線壓損計算(長春)

範例:長春管線與管件壓損計算

5.Darcy's formula:

$$h_{L(pipe)} = f\frac{L}{D}\frac{v^2}{2g} \qquad h_{L(Fitting\_and\_Valve)} = (f\frac{L}{D})\frac{v^2}{2g} = K\frac{v^2}{2g}$$

6.管線與管件壓損(m-Liquid):H_L= $H_{L(Pipe)}$ + $H_{L(Fitting + Valve)}$

$$h_L = (f\frac{L}{D}+K)\frac{v^2}{2g} = (0.0221x\frac{20}{(102.26/1000)}+3.667)\frac{(0.683)^2}{2x9.8}$$

→ H_L= 0.1902 m-Liquid

→ DP_{piping} = 0.1902m x 990 kg/m³ / 10000(cm²/m²)

= 0.019 (kg/cm²)

3.2 氣相管線壓損計算

氣體壓損計算可分為:

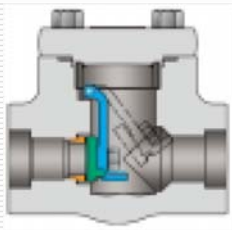
1.一般氣體(Vapors or Gases) : 參考速限表或實際允許壓降

2.微正壓氣體(Vapors or Gases) :儲槽排氣管線或儲槽連通管線

管線總壓降（含Condenser + 控制閥 + Scrubber + Line）
需桶槽操作壓力 (125 mmAq)

若小尺寸管線(≦1-1/2")需要使用逆止閥(Check Valve)時，必須(一定)要使用Swing Type Check Valve。液體重力流管線同此原則。

3.蒸氣(Steam): 參考速限表或實際允許壓降。



3.2 氣相管線壓損計算

氣體管線壓損計算:

1.計算管線的壓損:

1.1. Darcy rational relation for compressible flow :

$$\frac{\Delta P}{100\,ft} = \frac{0.000336\, f W^2 \overline{V}}{d^5}$$

Applied Process Design for Chem. & Petrochem. Plants, vol.1, p.102, eq.2-77

V : Specific volume of fluid, ft^3/lb

W : Flow rate, lb/h

d : Pipe I.D. , in

DP : Pressure drop, psi

3.2 氣相管線壓損計算

氣體管線壓損計算:

Applied Process Design for Chem. and Petrochem. Plants, v.1, 3rd ed., p.108

1.2 Sonic Conditions Limiting Flow of Gases and Vapors :

Vs = Fluid sonic or critical velocity

$$v_s = \left[\left(\frac{c_p}{c_v}\right)(32.2)\left(\frac{1544}{MW}\right)(460+t)\right]^{\frac{1}{2}} = 68.1\left[\left(\frac{c_p}{c_v}\right)\frac{P'}{\rho}\right]^{\frac{1}{2}},\ ft/\sec$$

Cp/Cv : Ratio of specific heats

ρ : density of vapor 氣體密度(kg/m^3)

壓縮流體於管內之最大流速

From ITT Technical Data

3.2 氣相管線壓損計算

2.氣體管線壓損查圖法:

Flow of Fluids through Valves, Fittings, and Pipe, 1988, p.3-18~3-21

範例: 600 psig(43.2kg/cm^{2}g)、850^{o}F (454.4^{o}C)的Steam，
Steam流量為30,000 lbs/hr(13,608 kg/hr) 經過4” Sch80
管線，管線長度為120米長，有10個90^o Elbow，1顆
手動閥Globe Valve。

2.1.查圖雷諾數(Re)與friction factor :

μ = 0.029 , cp **Re= 1,700,000**

W = 30,000 , lb/h **f = 0.017**

2.2.查圖管線100 ft壓損(psi)

V = 1.22 , lb/ft^3

W = 30,000 , lb/h

d = 3.826 , in

μ = 0.029 , cp

DP_{100} = 7.5, psi

From ITT Technical Data

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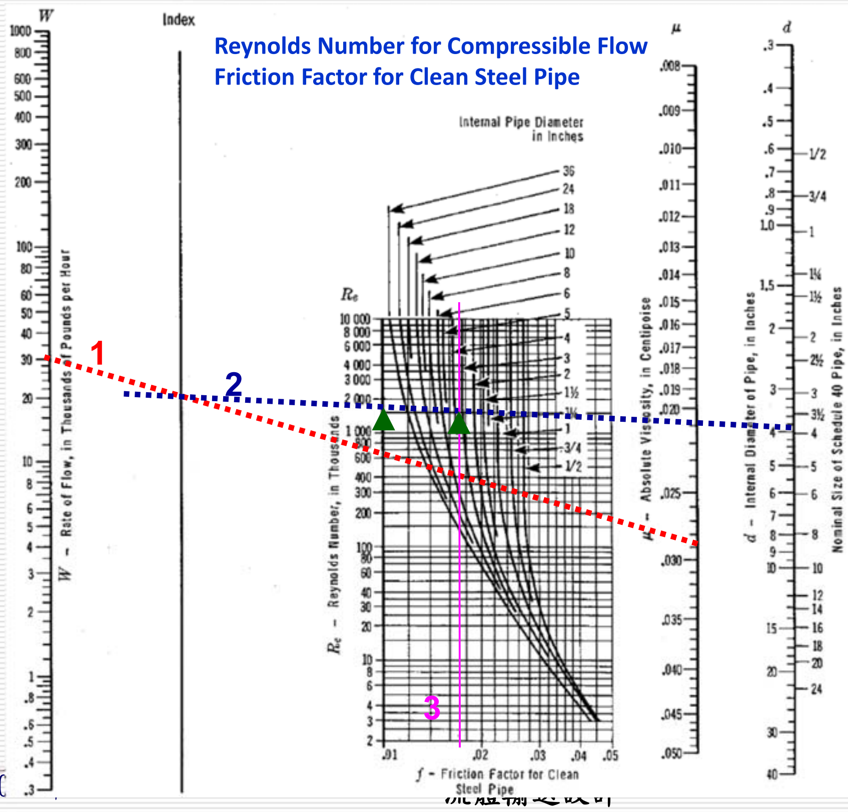
3.2 氣相管線壓損計算

2.1.查圖求雷諾數(Re)與friction factor :

Flow of Fluids through Valves, Fittings, and Pipe, 1988, p.3-18~3-21



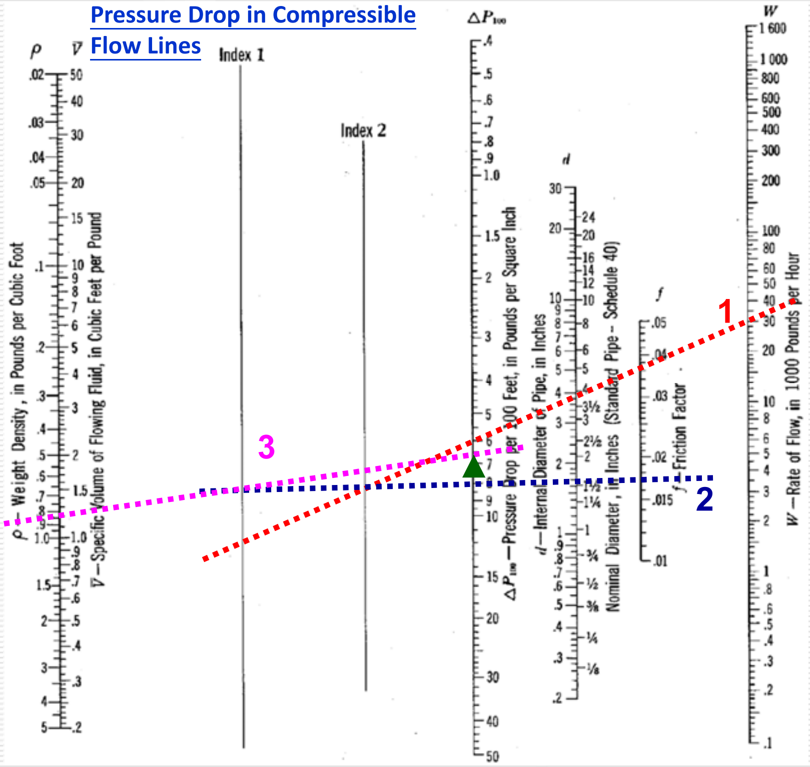
W = 30,000 , lb/h
μ = 0.029 , cp
d = 3.826 , in

Re= 1,700,000
f = 0.017

3.2 氣相管線壓損計算

2.2查圖管線100 ft壓損(psi)

Flow of Fluids through Valves, Fittings, and Pipe, 1988, p.3-18~3-21



W = 30,000 , lb/h

d = 3.826 , in

μ = 0.029 , cp

V = 1.22 , lb/ft^3

DP_{100} = 7.5, psi

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3.2 氣相管線壓損計算

3.長春氣相管線與管件壓損計算

Flow of Fluids through Valves, Fittings, and Pipe, 1988, p.3-18~3-21

3.1.計算雷諾數(Re)以確認流態(Laminar或Turbulent)

$$R_e = \frac{Dv\rho}{\mu_e} = \frac{(97.18)(39.4)(12.94)}{(0.029)} = 1{,}708{,}6478$$

3.2. Re>4000判定為Turbulent湍流

3.3. 相對粗糙度(Relative Roughness, ε/D)

ε/D=0.1/97.18=0.001029

$$f = \left(A - \frac{(B-A)^2}{C-2B+A}\right)^{-2} \qquad A = -2.0x\log\left(\frac{\varepsilon/D}{3.7} + \frac{12}{\mathrm{Re}}\right)$$

$$B = -2.0x\log\left(\frac{\varepsilon/D}{3.7} + \frac{2.51A}{\mathrm{Re}}\right) \qquad C = -2.0x\log\left(\frac{\varepsilon/D}{3.7} + \frac{2.51B}{\mathrm{Re}}\right)$$

3.2 氣相管線壓損計算

範例:長春管線與管件壓損計算

3.4.計算friction factor

$$A = -2.0x\log(\frac{\varepsilon / D}{3.7} + \frac{12}{\mathrm{Re}}) = -2.0x\log(\frac{0.001029}{3.7} + \frac{12}{17086478}) = 7.08991$$

$$B = -2.0x\log(\frac{\varepsilon / D}{3.7} + \frac{2.51A}{\mathrm{Re}}) = -2.0x\log(\frac{0.001029}{3.7} + \frac{2.51x7.08991}{1708647}) = 7.079624$$

$$C = -2.0x\log(\frac{\varepsilon / D}{3.7} + \frac{2.51B}{\mathrm{Re}}) = -2.0x\log(\frac{0.001029}{3.7} + \frac{2.51x7.079624}{1708647}) = 7.079669$$

$$f = (A - \frac{(B-A)^2}{C-2B+A})^{-2} = 0.01995$$

求出管線之 f = 0.0195

3.2 氣相管線壓損計算

範例:長春管線與管件壓損計算

3.5.求出管件(Fitting)之K(from Two-K Method)

3.6. Equivalent Length Le=KD/F=9.069x97.18/1000/0.01995

=44.18(m)

ELBOW	K1_fit	K2_fit	Qty	K
90°ANGLE :	K_1	K_∞		
long-radius, all types	800	0.2	10	2.523
VALVES :				
Globe, standard	1500	4	1	5.046
PIPE ENTRANCE, normal	160	0.5	1	0.500
PIPE EXIT	0	1	1	1.000
**** SUM ****				9.069

3.2 氣相管線壓損計算

範例:長春管線與管件壓損計算

3.7.Darcy rational relation for compressible flow :

$$\frac{\Delta P}{100\,ft} = \frac{0.000336 f W^2 \overline{V}}{d^5}$$

DP=0.000336x0.01995x(13607kg/hr*2.205)^2/(12.94 kg/m³ *0.06243)/(97.18/25.4)^5*(120+44.2)*0.03281/14.224

=3.4509 kg/cm²

1kg/hr=2.205 lb/hr , 1 kg/m³ = 0.06243 lb/ ft³

1m=3.281 ft, 1 kg/cm² = 14.22 lb/ in²

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3.2 氣相管線壓損計算

注意事項:

1. $DP/P_{upstream}$最好不要超過 10%，安全閥 Inlet pipe 不要超過 3%

2.如果管線壓損/來源壓力($DP/P_{upstream}$)超過下列值可能到達 Critical or sonic flow

2.1. 42% for sat'd steam

2.2. 47% for diatomic gases (e.g., H_2、N_2 、 O_2)

2.3. 45% for triatomic and higher MW gases, including hydrocarbon vapors and superheated steam

**真空蒸餾 Overhead vapor, Venting system, 長管線要特別注意

(CEP March 1991, p.36)

3.2 氣相管線壓損計算

注意事項:

3. Darcy Equation是針對不可壓縮流體發展出來的，但可壓縮流體在下列情況下使用，仍可得到一具合理精確度的計算結果:

3.1.計算所得的壓降(P_1-P_2)＜10%P_1（10%入口壓力），以入口或出口的條件作為計算的基準。

3.2.計算所得的壓降10%P_1＜(P_1-P_2)＜40%P_1，以入口和出口條件的平均值作為計算的基準。

3.3.計算所得的壓降(P_1-P_2)＞40%P_1 ，則必須以可壓縮流體來處理。對於長的可壓縮流體管線的流動就常碰到此種情況。

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3.2 氣相管線壓損計算

注意事項:

4. Noise Limit: *Guidelines for Eng. Design for Process Safety P.187*

保持流速於範圍內可避免過大的管線噪音

4.1. Liquids (液體): V≦ 9.144 m/s

4.2.Gases、Vapors (氣體)和兩相流:

$V(m/s)=122/\sqrt{density(kg/m^3)}$

(如果是兩相流為氣液混合的密度)

V: Velocity

3.3 兩相流管線壓損計算

兩相流水平管線的流態說明:

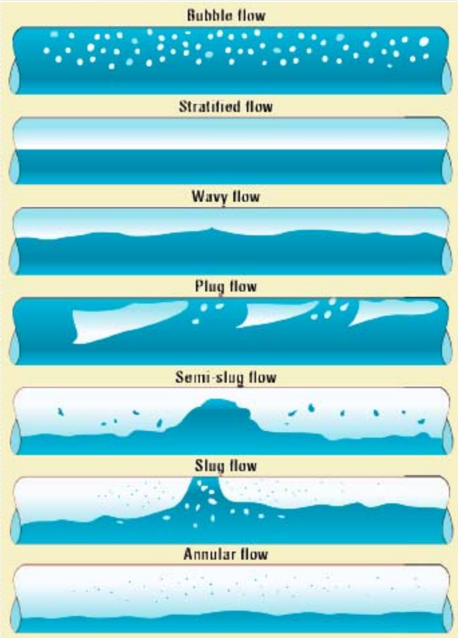


FIGURE 2. Horizontal flow patterns

TABLE 4-35 Characteristics Linear Velocities of Two-Phase Flow Regimes

Regime	Liquid Phase (ft/s)	Vapor Phase (ft/s)
Bubble or froth	5–15	0.5–2
Plug	2	< 4
Stratified	< 0.5	0.5–10
Wave	< 1.0	> 15
Slug	15 (but less than vapor velocity)	15–50
Annular	< 0.5	> 20
Dispersed, Spray, or Mist	Close to vapor velocity	> 200

Two-Phase Flow Patterns in Horizontal Pipe

Bubble Flow	氣泡流	氣泡沿管上部流動，氣液速度接近
Stratified Flow	層流	液體沿管底流動，氣体在液體上流動，形成平滑的氣液界面
Wavy Flow	波狀流	類似層流，氣體在較高速下流動，界面受氣體波動影響而擾動
Plug Flow	柱塞流	液體和氣體沿管上部交替呈柱塞狀流動
Slug Flow	柱狀流	氣體以較快速流動而周期性柱狀形成泡沫狀，比平均流速大很多的速度流動
Annular Flow	環狀流	液體於管壁成環狀流動，氣體於管中心高速流動

3.3 兩相流管線壓損計算

兩相流水平段管線設計原則:

1. 避免落在Slug流態，易造成水鎚效應。
2. 兩相流中液體可能被加速至氣體速度，這種加速效應將導致沖蝕-腐蝕現象發生，因此Erosion factor (沖蝕因子)一定要小於1(建議<0.6) 。
3. 氣液混合速度最好小於25m/s，不宜超過Noise Limit，速度不宜超過0.3倍的聲速(以氣液混合密度計算)。
4. 調整管徑大小使流態為Annular、Bubble(是否落在Annular或Bubble區則是由物性數據及氣液流量比所決定的)及Mist流態，改變管徑大小後(改變氣體速度)，通常會調整至Annular流態，接近Mist流態區域, 使其遠離slug流態。
5. Mist流態是氣液均勻混合(接近可壓縮氣體)

3.3 兩相流管線壓損計算

兩相流垂直管線的流態:

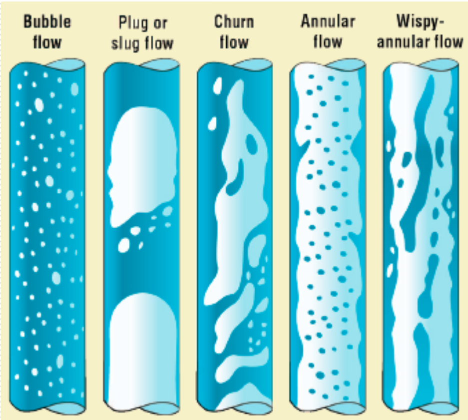


FIGURE 1. Vertical upflow patterns

Two-Phase Flow Patterns in Vertical Pipe

Bubble Flow	氣泡流	氣體呈氣泡分散於向上流動的液體中
Plug or Slug Flow	柱狀流	液體和氣體交替呈柱狀向上移動
Churn Flow	泡沫流	氣泡和液體混在一起，形成擾動紊亂的流態
Annular Flow	環狀流	液體以小於氣體的速度沿管壁向上移動，氣體於管中心向上移動，部份液體成液滴夾帶在氣體中
Wisp-Annular Flow	束狀環流	氣體中夾帶的液滴瞬間會結合在一起，呈束狀的液體於氣體的中心

3.3 兩相流管線壓損計算

管線垂直上升(Upflow)段設計原則:

1. 避免落在Slug流態，以及Churn流態(為發展為slug 流態的暫態情況，發生於上升段的入口處)。
2. 氣體速度需足夠才能進到欲設計的Annular流態。

3.3 兩相流管線壓損計算

管線垂直下降(Downflow)段設計原則:

1. 兩相流的液體在低流速下(<0.6m/s)多為Annular流態，當液體速度增加會過渡至Slug流態(避免落在Slug流態)，再增加液體速度則到Bubble流態。
2. 液體速度太快會造成管線Vibrate。

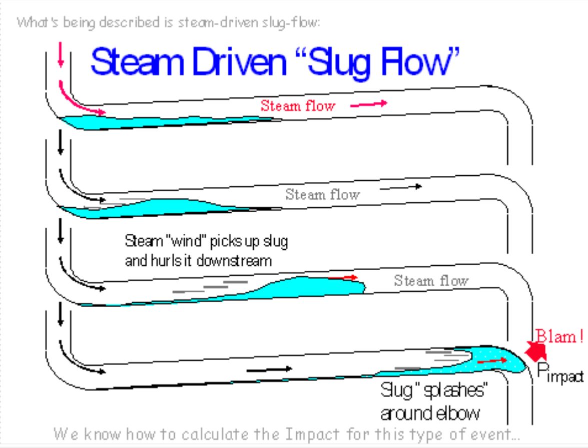
Modified Froude Number:

$Fr=(U_L2/(gD))^{0.5} \times (\rho_L/(\rho_L-\rho_G))^{0.5}$,

Fr ≦ 0.31 → Self-Venting

0.31 < Fr ≦ 1 → Pulse Flow

Fr > 1 → Pipe will vibrate



3.3 兩相流管線壓損計算

管線垂直下降(Downflow)段設計原則:

Modified Froude Number:

$Fr=(U_L2/(gD))^{0.5} \times (\rho_L/(\rho_L-\rho_G))^{0.5}$,

Fr ≦ 0.31 → Self-Venting

0.31 < Fr ≦ 1 → Pulse Flow

Fr > 1 → Pipe will vibrate

也常應用於重力流管線，特別是使用於蒸餾塔的內部(Down Pipe)，會設計讓氣體Self-Venting。氣體會脫離向下流動的液體使向下流動的液體不含氣體。

重力流管線估算一般重力流坡度要求1:100(0.57^0)

3.3 兩相流管線壓損計算

Modified Froude Number也是判斷重力流管線很好的依據:

Modified Froude Number:

也常應用於重力流管線，特別是使用於蒸餾塔的內部(Down Pipe)，會設計讓氣體Self-Venting。氣體會脫離向下流動的液體使向下流動的液體不含氣體。

重力流管線(密閉收集系統/污雨水系統):

1.一般重力流坡度(Slope)要求1:100(0.57^0)

2.管線尺寸≦200mm 以2倍液體流量設計

3.管線尺寸>200mm 以1.3倍液體流量設計

(CRC Press) Process Engineering and Design Using Visual Basic (2007)

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3.3 兩相流管線壓損計算

沖蝕腐蝕(Erosion-Corrosion):

$$Erosion\_Corrosion = \rho_M U_M^2 \leq 15{,}000$$

$$\rho_M = \frac{W_L + W_G}{\left(\frac{W_L}{\rho_L} + \frac{W_G}{\rho_G}\right)}$$

$$U_M = U_G + U_L$$

W_L, W_G =liquid (gas) flow rate, kg/h

ρ_L, ρ_G =liquid (gas) density, kg/m^3

U_L , U_G =Superficial Linear liquid (gas) velocity, m/s

A = inside cross-sectional area of pipe, m^2

$$U_M = \left(\frac{W_G}{3600\rho_G A} + \frac{W_L}{3600\rho_L A}\right)$$

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3.3 兩相流管線壓損計算

沖蝕腐蝕(Erosion-Corrosion):

依據不同的流態，液體在兩相流系統液體會被加速使速度接近或超過氣體的速度，高速度會導致沖蝕腐蝕現象，受腐蝕物質(Erosive Material)或沖蝕力(例如高速的液體)會加速腐蝕速率。

亦會受液體和氣體不同的化學物質本質，例如氧氣等

沖蝕因子是被用來決定混合密度與速度

$$Erosion\_Factor = \rho_M U_M^2 / 15{,}000$$

沖蝕因子≦1, it is OK! (建議<0.6)

沖蝕因子>1, erosion and corrosion may occur!

Chem. Eng. Prog., Nov., p.64 (1990) "Understand Two-Phase Flow in Process Piping"

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3.3 兩相流管線壓損計算

Flow regime map for horizontal two-phase flow:

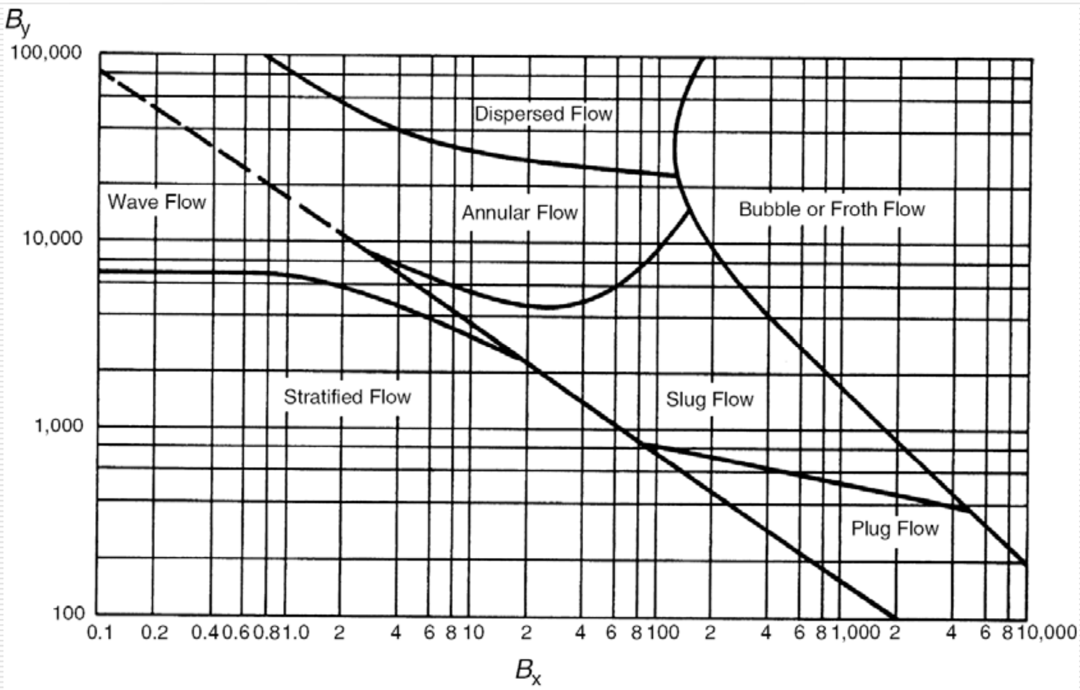

Figure 4-48a Flow patterns for horizontal two-phase flow. (Based on data from 1-, 2-, and 4-in. pipe). (By permission from Baker O., Oil Gas J.,Nov 10, 1958, p. 156.)

3.3 兩相流管線壓損計算

Flow regime map for horizontal two-phase flow:

Baker Parameter (確認Flow Patterns)

$$B_x = 531\left(\frac{W_L}{W_G}\right)\left\{\frac{(\rho_L \rho_G)^{0.5}}{\rho_L^{2/3}}\right\}\left\{\frac{\mu_L^{1/3}}{\sigma_L}\right\}$$

$$B_y = 2.16\left(\frac{W_G}{A}\right)\frac{1}{(\rho_L \rho_G)^{0.5}}$$

W_L, W_G =liquid (gas) flow rate, lb/h

ρ_L, ρ_G =liquid (gas) density, lb/ft^3

μ_L , μ_G =liquid (gas) viscosity, cp

σ_L =liquid surface tension, dyne/cm

A = inside cross-sectional area of pipe, ft^2

3.3 兩相流管線壓損計算

兩相流壓損計算式:

$$\Delta P_{TP} = \Delta P_G \Phi^2$$

Lockhart-Martinelli two- phase flow modulus,

$$X = \left[\frac{(\Delta P/\Delta L)_L}{(\Delta P/\Delta L)_G}\right]^{0.5} \qquad \Phi = f(x)$$

$(DP/DL)_L$ = 實際液體流經管線的壓損

$(DP/DL)_G$ =實際氣體流經管線的壓損

A = 管線的截面積(ft^2)

3.3 兩相流管線壓損計算

兩相流各流態的Φ值整理如下:

Type Flow	Equation for Φ_{GTT}
Froth or Bubble	$\Phi = \left[\dfrac{14.2X^{0.75}}{(W_L/A)^{0.1}}\right]$
Plug	$\Phi = \left[\dfrac{27.315X^{0.855}}{(W_L/A)^{0.17}}\right]$
Stratified	$\Phi = \left[\dfrac{15{,}400X}{(W_L/A)^{0.8}}\right]$
Slug	$\Phi = \left[\dfrac{1190X^{0.815}}{(W_L/A)^{0.5}}\right]$
Annular	$\Phi = (4.8 - 0.3125d)X^{(0.343-0.021d)}$

d is the inside diameter of the pipe in inches; for pipes 12 in. and larger, use d=10

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3.3 兩相流管線壓損計算

範例:兩相流管線與管件壓損計算

1.一股氣液流體管線長度約190米，評估此流體的流態與預估壓損

2. 管線為Sch40S

		Liquid	Gas
Flow W	kg/hr	454	1361
Density ρ	kg/m^3	1009	1.233
Viscosity μ	cP	1.0	0.077
Surface Tension σ	dyn/cm	15	

3.3 兩相流管線壓損計算

範例:兩相流管線與管件壓損計算

1.決定流體管線內可能的流態(計算Bx)

$$B_x = 531\left(\frac{W_L}{W_G}\right)\left\{\frac{(\rho_L \rho_G)^{0.5}}{\rho_L^{2/3}}\right\}\left\{\frac{\mu_L^{1/3}}{\sigma_L}\right\}$$

1 kg/cm³=0.06243 lb/ft³

$$B_x = 531\left(\frac{454}{1361}\right)\left\{\frac{[(1009*0.06243)(1.233*0.06243)]^{0.5}}{(1009*0.06243)^{2/3}}\right\}\left\{\frac{1}{15}\right\}$$

Bx = 1.62

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3.3 兩相流管線壓損計算

範例:兩相流管線與管件壓損計算

2.決定流體管線內可能的流態(計算By)

$$B_y = 2.16\left(\frac{W_G}{A}\right)\frac{1}{(\rho_L \rho_G)^{0.5}}$$

1 kg/hr = 2.205 lb/hr

Sch 40S Pipe ID = 77.93mm = 3.068 inch = 0.2556 ft

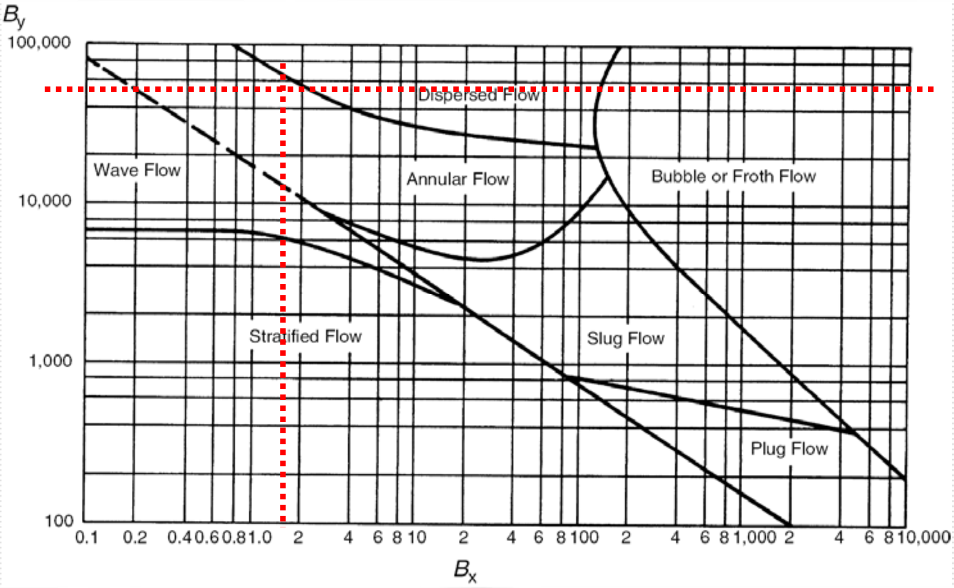
$$A = \frac{\pi D^2}{4} = \frac{\pi (0.2556)^2}{4} = 0.0513\,ft^2$$

$$B_y = 2.16(1361*2.204/0.0513)\frac{1}{\sqrt{(1009*0.0624)(1.233*0.0624)}}$$

By= 57364

3.3 兩相流管線壓損計算

3.查下圖流態可能為Annular Flow，因許多未訂與未知的條件，所以流態或許為Wave或Dispersed



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3.3 兩相流管線壓損計算

4.計算液體壓損:

4.1.計算雷諾數(Re)以確認流態(Laminar或Turbulent)

$$R_e = \frac{Dv\rho}{\mu_e} = \frac{(77.93)(0.026)(1009)}{(1.0)} = 2044$$

$$f = \frac{64}{R_e} = \frac{64}{2044} = 0.0313$$

4.2. 求出管件(Fitting)之K(from Two-K Method)

ELBOW	K1_fit	K2_fit	Qty	K
90°ANGLE :	K_1	K_∞		
long-radius, all types	800	0.2	3	1.970
VALVES :				
Gate, ball, plug :				
full line size, B=1.0	300	0.1	1	0.279
PIPE ENTRANCE, normal	160	0.5	1	0.578
PIPE EXIT	0	1	1	1.000
**** SUM ****				3.827

3.3 兩相流管線壓損計算

4.3.Darcy's formula:

$$h_{L(pipe)} = f\frac{L}{D}\frac{v^2}{2g} \qquad h_{L(Fitting\_and\_Valve)} = (f\frac{L}{D})\frac{v^2}{2g} = K\frac{v^2}{2g}$$

4.4.管線與管件壓損(m-Liquid):H_L= $H_{L(Pipe)}$ + $H_{L(Fitting + Valve)}$

$$h_L = (f\frac{L}{D} + K)\frac{v^2}{2g} = (0.0313x\frac{190}{(77.93/1000)} + 3.827)\frac{(0.026)^2}{2x9.8}$$

→ **H_L= 0.00028 m-Liquid**

→ **DP_{piping} = 0.00028m x 1009 kg/m³ / 10000(cm²/m²)**

= 0.00028 (kg/cm²)

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3.3 兩相流管線壓損計算

5.計算氣體壓損:

5.1.計算雷諾數(Re)以確認流態(Laminar或Turbulent)

$$R_e = \frac{Dv\rho}{\mu_e} = \frac{(77.93)(64.3)(1.233)}{(0.00127)} = 4,864,912$$

5.2. Re>4000判定為Turbulent湍流

5.3. 相對粗糙度(Relative Roughness, ε/D)

ε/D=0.1/77.93=0.00128

From ITT Technical Data

3.3 兩相流管線壓損計算

5.4.計算f (Friction Factor)值

$$A = -2.0x\log(\frac{\varepsilon/D}{3.7} + \frac{12}{\mathrm{Re}}) = -2.0x\log(\frac{0.00128}{3.7} + \frac{12}{4864912}) = 6.91366$$

$$B = -2.0x\log(\frac{\varepsilon/D}{3.7} + \frac{2.51A}{\mathrm{Re}}) = -2.0x\log(\frac{0.00128}{3.7} + \frac{2.51x7.08991}{4864912}) = 6.91092$$

$$C = -2.0x\log(\frac{\varepsilon/D}{3.7} + \frac{2.51B}{\mathrm{Re}}) = -2.0x\log(\frac{0.00128}{3.7} + \frac{2.51x7.079624}{4864912}) = 6.91092$$

$$f = (A - \frac{(B-A)^2}{C-2B+A})^{-2} = 0.02094$$

求出管線之friction factor = 0.0209

3.3 兩相流管線壓損計算

5.5.計算氣體之管件(Fitting)之K (from Two-K Method)值

ELBOW	K1_fit	K2_fit	Qty	K
90°ANGLE :	K_1	K_∞		
long-radius, all types	800	0.2	3	0.796
VALVES :				
Gate, ball, plug :				
full line size, B=1.0	300	0.1	1	0.133
PIPE ENTRANCE, normal	160	0.5	1	0.500
PIPE EXIT	0	1	1	1.000
**** SUM ****				2.429

5.6. Equivalent Length

Le=KD/F=2.429x77.93/1000/0.02094

=9.04(m)

3.3 兩相流管線壓損計算

5.7.計算氣體的管線壓損:

Darcy rational relation for compressible flow :

$$\frac{\Delta P}{100\,ft} = \frac{0.000336\,f W^2 \overline{V}}{d^5}$$

V = Specific volume of fluid, ft³/lb

DP=0.000336x0.02094x(1361kg/hr*2.205)^2/(1.233 kg/m³ *0.06243)/(77.93/25.4)^5*(190+9.04)*0.03281/14.224

=1.390 kg/cm²

1kg/hr=2.205 lb/hr , 1 kg/m³ = 0.06243 lb/ ft³

1m=3.281 ft, 1 kg/cm² = 14.22 lb/ in²

3.3 兩相流管線壓損計算

5.8.計算X值:

Lockhart-Martinelli two- phase flow modulus,

$$X = \left[\frac{(\Delta P/\Delta L)_L}{(\Delta P/\Delta L)_G}\right]^{0.5} \qquad \Phi = f(x)$$

$$X = \left[\frac{0.00028}{1.390}\right]^{0.5} = 0.014$$

$(DP/DL)_L$ = 實際液體流經管線的壓損
$(DP/DL)_G$ =實際氣體流經管線的壓損
A = 管線的截面積(ft^2)

3.3 兩相流管線壓損計算

5.9. 確定兩相流流態為Annular Flow計算的Φ_{GTT}值:

$$\Phi = (4.8 - 0.3125d)X^{(0.343-0.021d)}$$

$$\Phi = (4.8 - 0.3125*(77.93/25.4))*0.014*^{(0.343-0.021*(77.93/25.4))}$$

$\Phi = 1.170$

Note: d is the inside diameter of the pipe in inches; for pipes 12 in. and larger, use d=10

$$\Delta P_{TP} = \Delta P_G \Phi^2$$

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3.3 兩相流管線壓損計算

5.10.計算兩相流之壓損:

$$\Delta P_{TP} = \Delta P_G \Phi^2$$

$DP_{TP} = 1.390 \times 1.17^2$

$DP_{TP} = 1.903\ kg/cm^2$

管線設計參考

Piping Design Guide Line(參考1/3):

可用來Sizing和Rating管線的水力計算

建議的流速與壓損限制如下表:

Service	Maximum dP/100m $(kg/cm^2)/(100m)$	Maximum Velocity m/s
Vacuum Service Vapor Lines	0.13 P	
Compressor Suction (last suction vessel to compressor)	$0.037(P)^{1/2}$ or 0.23	
Reciprocating		$80/(MW)^{1/2}$
Centrifugal		$120/(\rho_{Vap})^{1/2}$
Compressor Discharge (compressor to liquid feed mix point, etc.)	$0.037(P)^{1/2}$ or 0.23	
Reciprocating		$80/(MW)^{1/2}$
Centrifugal		$120/(\rho_{Vap})^{1/2}$
Vapor Phase Reactor Circuits	0.45	$120/(\rho_{Mix})^{1/2}$

Where: P: absolute pressure, $kg/cm^2(a)$ MW: molecular weight

ρ_{Liq}, ρ_{Vap}: liquid and vapor density, kg/m^3

ρ_{Mixp}: mixed density $((w_{Liq} + w_{Vap})/(Q_{Liq} + Q_{Vap}))$,, kg/m^3

w_{Liq}, w_{Vap}: liquid and vapor mass flow, kg/hr

Q_{Liq}, Q_{Vap}: liquid and vapor volumetric flow, m^3 /kg

管線設計參考

Piping Design Guide Line(參考2/3):

Service	Maximum dP/100m $(kg/cm^2)/(100m)$	Maximum Velocity m/s
Pump Suction		
Bubble Point Liquid	0.035	
Subcooled	0.080	
(liquid subcooled at least 28℃ or vapor pressureat least $1.8kg/cm^2$ below system pressure)		
Pump Discharge	0.35	3.6
Gravity Feed	0.035	
General Service (based on available control valve ΔP)		
$\Delta P \geqq 3.5 kg/cm^2$	0.45	Liquid: 3.6
$1.8 \leqq \Delta P < 3.5 kg/cm^2$	0.35	Vapor: $120/(\rho_{Vap})^{1/2}$
$\Delta P < 1.8 kg/cm^2$	0.12	Mixed: $120/(\rho_{Mix})^{1/2}$
Liquid to Non-Pumped Reboiler	0.045	
Thermosiphon Reboiler Return to Column	0.035	$1.5((\rho_{Liq}-\rho_{Vap})/(\rho_{Vap})^{1/2})$
Forced Circulation Reboiler Return	0.28	$120/(\rho_{Mix})^{1/2}$

管線設計參考

Piping Design Guide Line(參考3/3):

Service	Maximum dP/100m (kg/cm^2)/(100m)	Maximum Velocity m/s
All Other Two-Phase Lines		$120/(\rho_{Mix})^{1/2}$
P≧22kg/cm^2(a)	0.45	
12≦P＜22kg/cm^2(a)	0.23	
1.0≦P＜12kg/cm^2(a)	0.12	
0.25≦P＜1.0kg/cm^2(a)	0.035	
P＜0.25kg/cm^2(a)	0.13P	
Column Overhead Vapor		
P≧21kg/cm^2(a)	0.3	
10.5≦P＜21kg/cm^2(a)	0.15	
1.0≦P＜10.5kg/cm^2(a)	0.07	
0.25≦P＜1.0kg/cm^2(a)	0.035	
P＜0.25kg/cm^2(a)	0.13P	
Condenser Outlet	0.058	
Cooling Water	0.30	
Q＜160m^3/hr		---
160≦Q＜640m^3/hr		2.5
640≦Q＜1600m^3/hr		3
Q≧1600m^3/hr		3.5
Steam	0.25	
Saturated		30
Superheated(at least 11℃)		75

管線設計參考

Piping Design Guide Line(參考1/2):

Table 2.5 Sizing Criteria for Carbon Steel Liquid Lines

Description	Velocity (m/sec)			ΔP (kPa/100 m)
Nominal pipe size (mm)	≤50	80–250	≥300	
Hydrocarbon, organic liquids				
Pump suction				
Nonboiling	0.3–0.9	0.9–1.8	1.2–2.4	5–20
Boiling	0.3–0.6	0.6–1.2	0.9–1.8	5–10
Pump discharge				
Short length	1.2–2.7	1.5–3.0	2.4–3.7	24–60
Long length	0.6–0.9	0.9–1.5	1.2–2.1	24–35
		For all sizes		
Liquid from condenser		0.9–1.8		5–10
Refrigeration lines		0.6–1.2		5–9
Gravity flow lines		0.9–2.5		9, Maximum
Reboiler inlet		0.3–1.2		2–6
Liquid feed to tower		1.2–1.8		6–10
Water				
General service	0.6–1.5	1.5–3.7	3.0–4.9	10–45
Pump suction		0.6–2.1		5–11
Pump discharge		0.6–4.9		10–45
Boiler feed		2.5–4.5		
Cooling water		3.6–4.8		10–45
Caustic solution		1.2–1.6		
Viscous Oil				
Pump suction	0.2 Maximum	0.2–0.3	0.2–0.3	NPSH
Pump discharge	0.3–0.9	0.9–1.5		

(CRC Press) Process Engineering and Design Using Visual Basic (2007)

管線設計參考

Piping Design Guide Line(參考2/2):

Table 2.6 Sizing Criteria for Carbon Steel Vapor Lines

Description	Maximum ρv^2 (kg/[m·sec²])	Maximum Velocity (m/sec)	Maximum *DP* (kPa/100 m)
Continuous Operation			
p < 2,000 kPaG	6,000		4–8
2,000 < p < 5,000	7,500		8–11
5,000 < p < 8,000	10,000		11–20
p > 8,000 kPaG	15,000		20–27
Intermittent Operation			
p < 2,000 kPaG	7,500		
2,000 < p < 5,000	9,500		
5,000 < p < 8,000	12,500		
p > 8,000 kPaG	20,000		
Compressor suction lines		20	3–7
Compressor Discharge Lines			
p < 500 kPaG		30	4
500 < p < 1,000		30	7
1,000 < p < 5,000		30	10
p > 5,000		30	12

(CRC Press) Process Engineering and Design Using Visual Basic (2007)

管線設計參考

最適化的管線尺寸:

長途經濟管徑(Economic Pipe Diameter):

NPS (in.)	Material	Sch.	OD (mm)	管厚 (mm)	每米管重 (kg/m)	管線費用 NTD	管線單價 NTD/6M	管線單價 NTD/kg	管損 kg/cm²
6	CS	40S	168.3	7.11	28.26	3,287,000	5,190	30.6	2.83
	304 SS	10S	168.3	3.40	13.84	5,826,667	9,200	110.8	2.28
		40S	168.3	7.11	28.26	12,666,667	20,000	118.0	2.83
8	CS	40S	219.1	8.18	42.56	4,940,000	7,800	30.5	0.71
	304 SS	10S	219.1	3.76	19.96	8,866,667	14,000	116.9	0.58
		40S	219.1	8.18	42.56	18,303,333	28,900	113.2	0.58

Note 1.長途管線長度約3800M, Note 2.管線單價為近日採購的實際金額
Note 3.管線管損是以80M3/hr估算(目前實際輸送量)

CS/304 SS長途管線管線費用(不含配管/管配件/搭架費)比較

管線設計參考與注意事項

非極性(不導電)流體流速之限制(避免靜電的危害)

- ✓ Conductivity < 100 pS/m
- ✓ Resistivity > 10 W m
- ✓ Benzene, Toluene , Xylene, Cyclohexane, Hexane etc
- ✓ $V_{max} < (0.21/D)^{0.5}$; D = Pipe ID (m)

參考速限如下:

NPS	in	3/4	1	1-1/2	2	2-1/2	3	4	6	8	10
Vmax	m/s	3.0	2.7	2.2	2.0	1.7	1.6	1.4	1.1	1.0	0.9

From : Avoid Static Ingnition Hazards in Chemical Operations, By Laurence G.Britton

管線設計參考與注意事項

非極性(不導電)流體流速之限制(避免靜電的危害)

Nonconductive Liquids (κ < 100 pS/m)

Liquid	Electrical Conductivity κ (pS/m)	ε_r	Charge Relaxation Time $\tau = \varepsilon_0\varepsilon_r / \kappa$ (s)
anisole (methyl phenyl ether)	10	4.33	3.8
benzene (purified)	5 x 10^{-3}	2.3	~100 (dissipation)
biphenyl (solid: less than 69°C)	0.17		n/a
bromine (17.2°C)	13		
butyl stearate	21	3.111	1.3
caprylic acid (octanoic acid)	<37	2.45	> 0.58
carbon disulfide (1°C)	7.8 x 10^{-4}	2.6	~100 (dissipation)
carbon tetrachloride	4 x 10^{-4}	2.238	~100 (dissipation)
chlorine (-70°C)	<0.01		
cyclohexane	<2	2.0	>8.8
decalin	6	2.18	3.2
dichlorosilane			
diesel oil (purified)	~0.1	~2	~100 (dissipation)
diethyl ether	30	4.6	1.4
1,4-dioxane (diethylene oxide)	0.1	2.2	~100 (dissipation)
ethyl benzene	30	2.3	0.68
gasoline (straight run)	~0.1	~2	~100 (dissipation)
gasoline (unleaded)	<50		
	(varies)		

Nonconductive Liquids (κ < 100 pS/m)

Liquid	Electrical Conductivity κ (pS/m)	ε_r	Charge Relaxation Time $\tau = \varepsilon_0\varepsilon_r / \kappa$ (s)
heptane (purified)	3 x 10^{-2}	2.0	~100 (dissipation)
hexane (purified)	1 x 10^{-5}	1.90	~100 (dissipation)
hexamethyldisilazane	29		
jet fuel A and A-1	0.01-50	2.2	0.39 - 100
jet fuel B	0.01-50	2.2	0.39 - 100
kerosene	1-50	2.2	0.39 - 19
methyl methacrylate			
methyl tertiary butyl ether (MTBE)			
naphthas	various	~2	~100 (dissipation)
pentachlorodiphenyl	0.8	5.06	~100 (dissipation)
pentachloroethane	<100	3.83	>0.3
SiH fluid (Y-10354)	2.5		
silicon tetrachloride			
stearic acid (80°C)	<40		
styrene monomer	10	2.43	2.2
toluene	<1	2.38	~100 (dissipation)
trichlorosilaneturpentine	22		
iso-valeric acid	<40	2.64	>0.58
xylene	0.1	2.38	~100 (dissipation)

管線設計參考與注意事項

管末噴出的流速限制 (1/2)

✓對於液體管末噴出目的地可能會有空氣時(例如:53加崙桶灌充、槽車灌充等)，一定要遵守流速限制規範，管末出口會成一道水流狀通過空氣落入槽內，會形成小液滴和霧狀微粒。液體自由落下通過空氣會產生靜電，並會導致火花放電而可能點燃易燃液體。

✓流速必須限制在1 m/s以下直到汲液管(Dip Pipe)被淹沒至約150mm高為止，才可調高流速至公式限制的流速，但最大仍不可大於7 m/s。

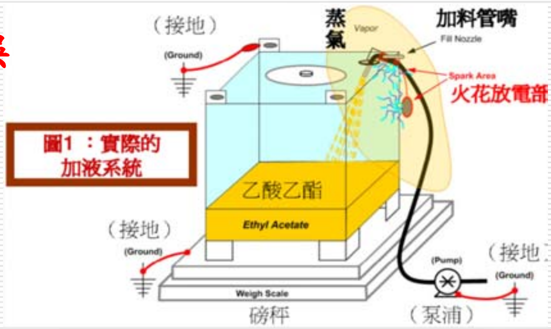
V x D ≦ 0.5 (m2/s)
V: Velocity in the down spout (m/s)
D: Inside diameter of the down spout (m)

管線設計參考與注意事項

管末噴出的流速限制 (2/2)

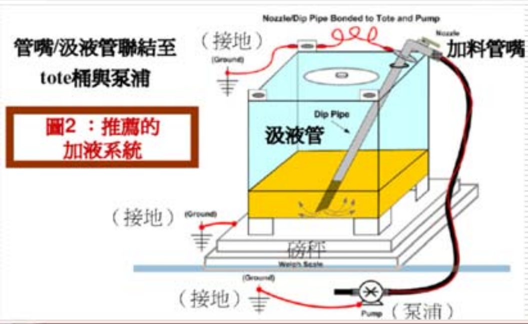
✓本質安全是使用汲液管(Dip Pipe) 或由底部入料。

錯誤



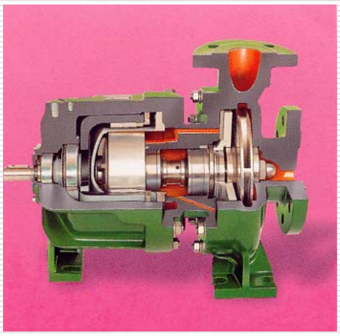
錯誤結果

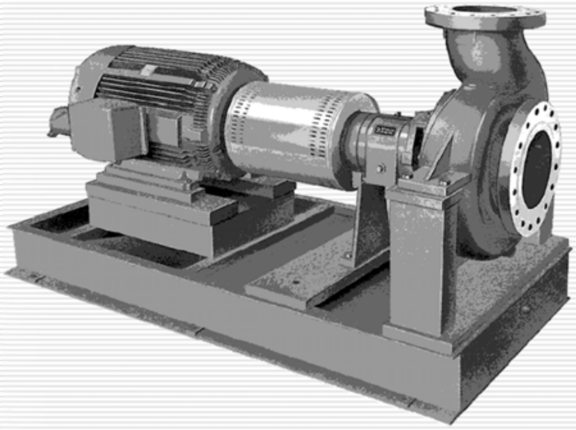
正確



Process Safety Beacon for Center for ChemicalProcess Safety, January 2009

4.泵浦的揚程(Head)與(NPSHa)計算





4.1 泵浦的NPSHa計算

NPSHa (available):為製程系統賦予泵浦的有效NPSH。

$$NPSHa = \frac{10 \times (P_1 - P_{vp} - \Delta P_f)}{SG} + (Z_1 - Z_{inlet})$$

NPSH (Net Positive Suction Head) :淨正吸入水頭 (m)

P_1:來源壓力(kg/cm²g)

P_{vp}:流體蒸氣壓(kg/cm²)

ΔP_f:壓損，包含管線/儀表/過濾器等(kg/cm²)

SG:比重 Relative Density (Specific Gravity)

Z_1:來源設備最低液位高程(m)

Z_{inlet}:泵浦入口中心之位高(m)

4.1 泵浦的NPSHa計算

範例: 請計算P-259之NPSHa、Head(揚程)、控制閥壓損(DP_{cv})

V-259設備高程 3米高，設備之操作壓力0.1kg/cm²g，流體為38℃的水，其水於38℃下的蒸氣壓(Vapor Pressure)為0.068kg/cm²，密度(Density)為990 kg/m³。

P-259泵浦入口中心高程0.75米，泵浦入口有一組Y型過濾器，泵浦入口為4” S40S管線，管線長度為20米長，有四個90º Elbow，一個Tee (Used as elbow)，手動閥Gate Valve 1個。

P-259泵浦的操作流量為20,000 kg/hr，流體的黏度(Viscosity)為0.697 cP。

P-259泵浦出口為3” S40S管線，管線長度為60米長，有10個90º Elbow，手動閥Gate Valve與Swing Check Valve 各1個。

P-259泵浦出口泵至TK-259，TK-259操作壓力0.01kg/cm2g， TK-259可能最高的高程為15m，有一組流孔板流量計(FC-2591) 、控制閥(FV-2591)與熱交換器(H-259)計算壓損為0.25 kg/cm2。

4.1 泵浦的NPSHa計算

NPSHa結果計算如下:

P_1:0.1 (kg/cm²g)

P_{vp}:0.068 (kg/cm²)

ΔP_f:管線(0.068) /Y型過濾器(0.035) (kg/cm²)

SG:比重 Relative Density (Specific Gravity)

Z_1:高程 3 (m)

Z_{inlet}:泵浦入口位高 0.75 (m)

$$NPSHa = \frac{10\times(P_1 - P_{vp} - \Delta P_f)}{SG} + (Z_1 - Z_{inlet})$$

$$NPSHa = \frac{10\times((0.1+1.033)-0.068-(0.035+0.032))}{0.99} + (3-0.75)$$

$$NPSHa = 12.33(m)$$

4.1 泵浦的NPSHa計算

一般NPSHr 須同時滿足 :(各公司應該各有自已的基準)

(1) < NPSHa – 0.5m ; (2) < NPSHa x 70%

- ✓ 泵浦在操作時流體於泵浦的吸入端必需使流體保持於全液體的狀態 。
- ✓ 若NPSHa 不足時，流體易部分蒸發，造成 Cavitation (氣蝕)、出口壓力會不穩定及產生振動等現象，流量、揚程與效率更會大幅的降低，受Cavitation現象泵殼(Casing)與葉片(Impeller)會損壞，使泵浦壽命因此縮短。

4.1 泵浦的NPSHa計算

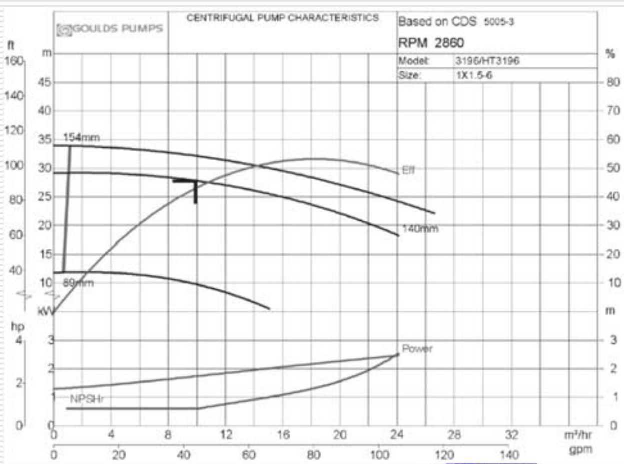
選擇泵浦時必須檢視泵浦廠商所提供的NPSHr是否滿足規範:(各公司應該各有自已的基準)

(1) < NPSHa – 0.5m ; (2) < NPSHa X 70%

計算範例:NPSHa=12.33m 。

(1) NPSHr < 12.33-0.5 = 11.03m ;

(2) NPSHr < 12.33 x 70% = 8.63m



4.1 泵浦的NPSHa計算

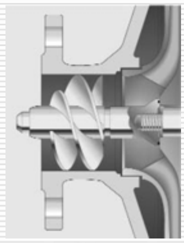
當預防氣蝕的發生其方法:

如何提高NPSHa：

- ✓ 提高吸入端設備內液位高度或提高設備高程 (Z_1)
- ✓ 增大管線尺寸降低管線壓損(ΔP_f)
- ✓ 冷卻製程流體降低流體蒸氣壓 (P_{vp})

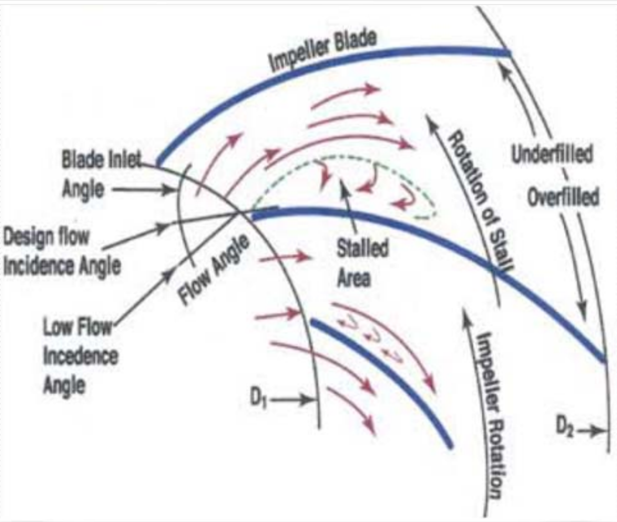
降低NPSHr ：

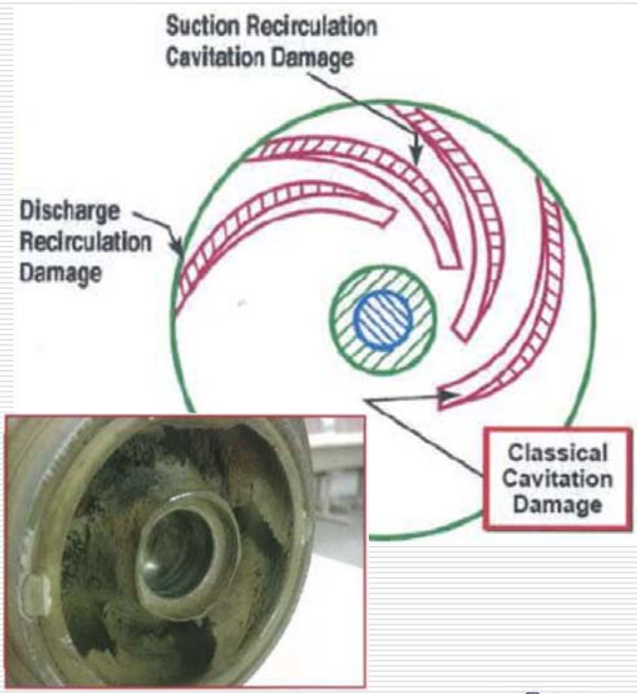
- ✓ 提高吸入端設備內降低馬達轉速
- ✓ 採用雙吸式葉片 (Double Suction Pump)
- ✓ 採用Inducer(可降低35~50%)



泵浦的氣蝕現象(Cavitation)

氣蝕現象(Cavitation)的判斷:





泵浦開車時的應用

泵浦現場TroubleShooting常用(簡單又好用)(1/2)

1 知道泵浦flow, density, head, efficiency算Power

Q	flow capacity	20	m^3/h	
D	density of fluid	499	kg/m^3	
H	differential head	350	m	
η	pump efficiency	34	%	
Ph	hydraulic Power	9.52	kw	$P_h = q\,\rho\,g\,h / 3.6x10^6$
		12.94	BHP	
Ps	shaft power	28.00	kw	Ph / η
		38.06	BHP	

2 知道泵浦的AMP算Power

A	amp	62.5	A	A:實際量測電流
Vi	voltage	440	V	Vi:馬達電壓
η_m	motor efficiency	85	%	$\eta_{m:}$馬達效率一般:80~90%
Ps	shaft power	40.49	kw	

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泵浦開車時的應用

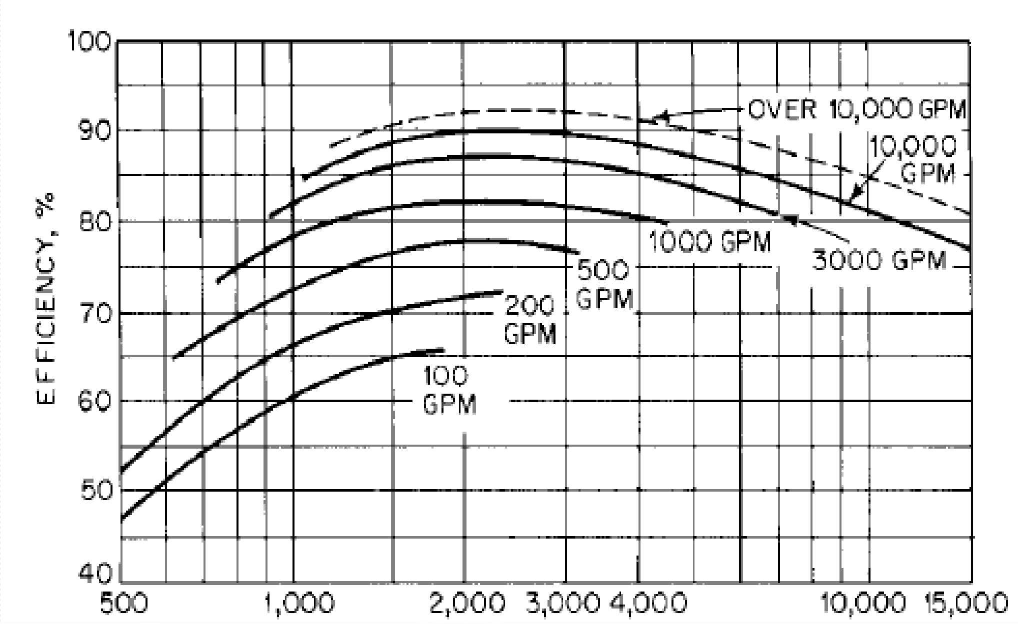
泵浦現場TroubleShooting常用(簡單又好用)(2/2)

3 知道泵浦flow, density, head, Amp算efficiency

Q	flow capacity	20	m^3/h	
D	density of fluid	499	kg/m^3	
H	differential head	350	m	
Ph	hydraulic Power	9.52	kw	$P_h = q\,\rho\,g\,h / 3.6x10^6$
A	amp	62.5	A	A:實際量測電流
Vi	voltage	440	V	Vi:馬達電壓
η_m	motor efficiency	85	%	$\eta_{m:}$馬達效率一般:80~90%
Ps	shaft power	40.49	kw	
η	pump efficiency	23.5	%	

泵浦開車時的應用

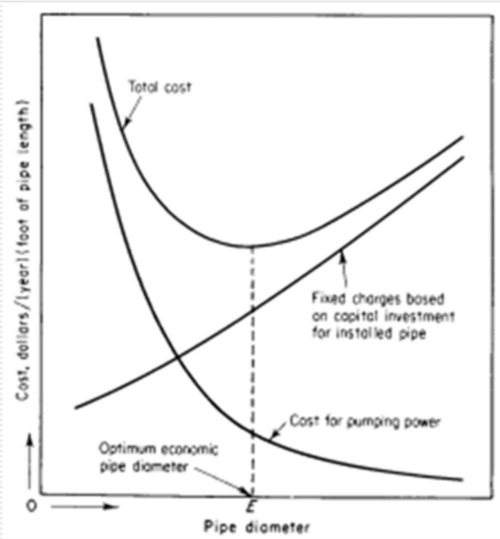
市面上一般泵浦效率值



泵浦與管線

經濟管徑(Economic Pipe Diameter)

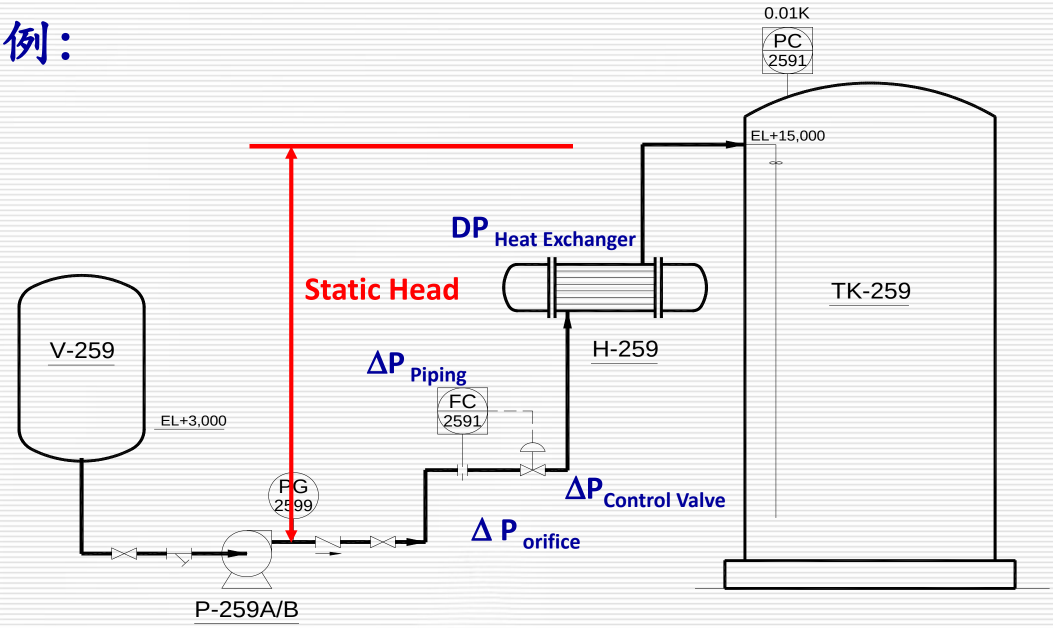
對於一泵浦的迴路，管線的管徑愈大則摩擦阻力愈小，泵浦需要的揚程愈小，也就是操作成本愈低，但設置的固定成本相對愈高；反之亦然，因此泵浦的吸入端及排出端管線尺寸決定於一平衡的最佳值（optimum），即固定成本和操作成本的和為最低的管徑。



Determination of optimum economic pipe diameter for constant mass-throughput rate

4.3泵浦的揚程(Head)

圖例:



4.3泵浦的揚程(Head)計算

1. Equipment Pressure :以可能之最高操作壓力為基準
2. Feed Nozzle Height:可能之最高操作液位為基準
3. Static Head:由2項算得
4. Friction Loss (壓損):

   Piping、Filter、Static Mixer、
   Flowmeter、Heat Exchanger、
   Equipment、Distributor等
5. Control Valve Pressure Drop
6. Pump Discharge Pressure

   = 1. + 3. + 4. + 5.

4.4泵浦的節能

既有泵浦需要節能原因:

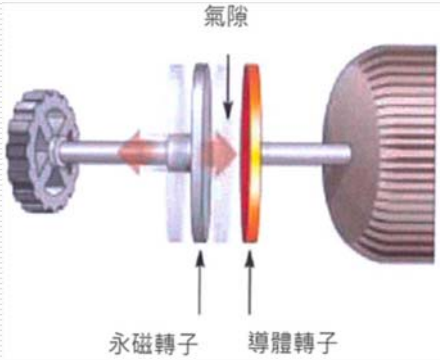
1. 更改操作條件,使揚程或流量變化
2. 製程特殊程序步驟時:短時間的高揚程或高流量
3. 設計預估的揚程偏高,尤其冷卻水泵浦

既有泵浦節能方法:

1. 修改葉輪尺寸(Impeller)
2. 調降轉速:

✓ 採用磁力聯軸器

✓ 採用變頻馬達



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4.4泵浦的節能

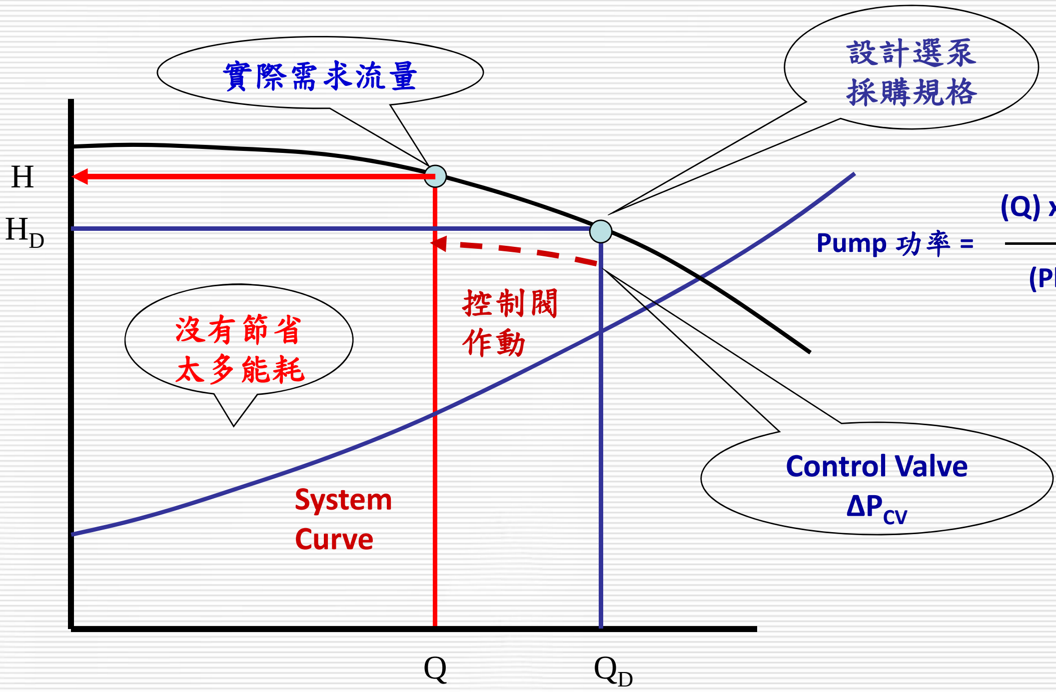
泵浦節能有三種方式

1)修改泵浦的葉片, 2)變頻馬達,3)使用永磁偶合器之比較如下:

項 次(節能方式)	更換葉片	變頻馬達	磁力偶合器
投資成本	中	高	低
工作原理	修改葉片	改變馬達電壓頻率轉速	無機械連接,永磁轉子與銅導體轉子氣隙改變泵浦轉速
流量/壓力調節	單一性能曲線,無調整空間	可(調整頻率)	可(調整泵浦轉速)
節能效果	普通(單一性能曲線)	優(可調整至最佳操作點)	優(可調整至最佳操作點)
修改工事	需要泵浦拆離,重新更換和組裝葉片	馬達的電纜線需重新鋪設 佔用MCC空間大	利用既有coupling空間,可不修改馬達基座
NPSHr	會升高(差)	會降低(優)	會降低(優)
BEP效率	下降	不變	不變
設備使用壽命	>20 年	>20 年	>20 年
馬達-泵軸對準	馬達與設備為機械連結,需精密校準,振動機率大	馬達與設備為機械連結,需精密校準,振動機率大	無機械連結,容忍對心不準度,減少振動,提升軸承與密封壽命

4.4泵浦的節能

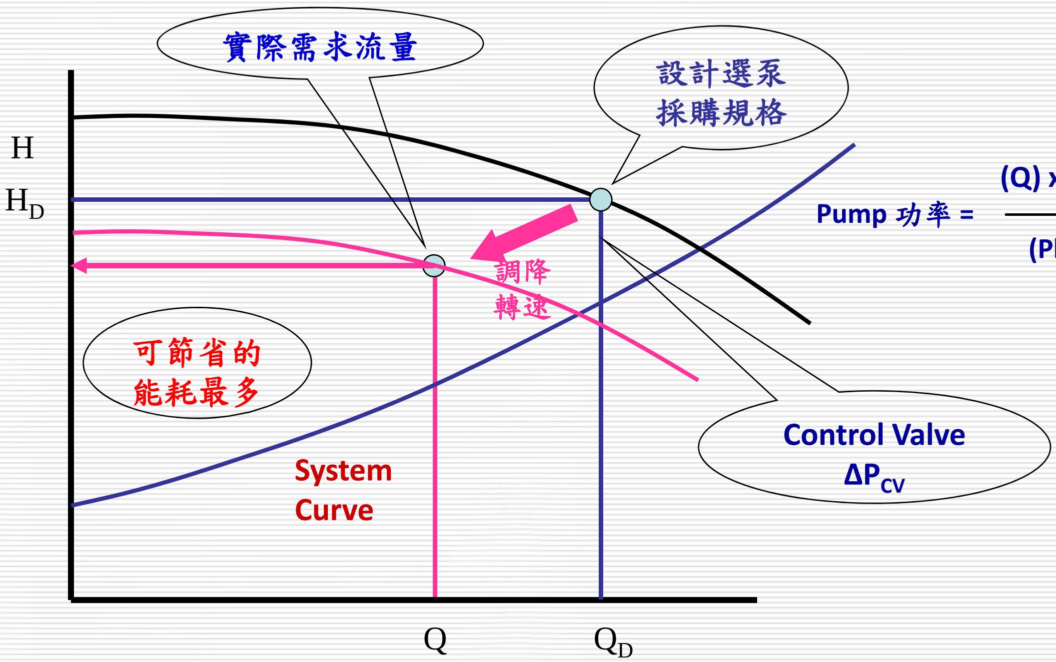
一般泵浦的操作模式:



$$\text{Pump 功率} = \frac{(Q) \times (H) \times (Sp.Gr.)}{(Ph) \times (Eff.)}$$

4.4泵浦的節能

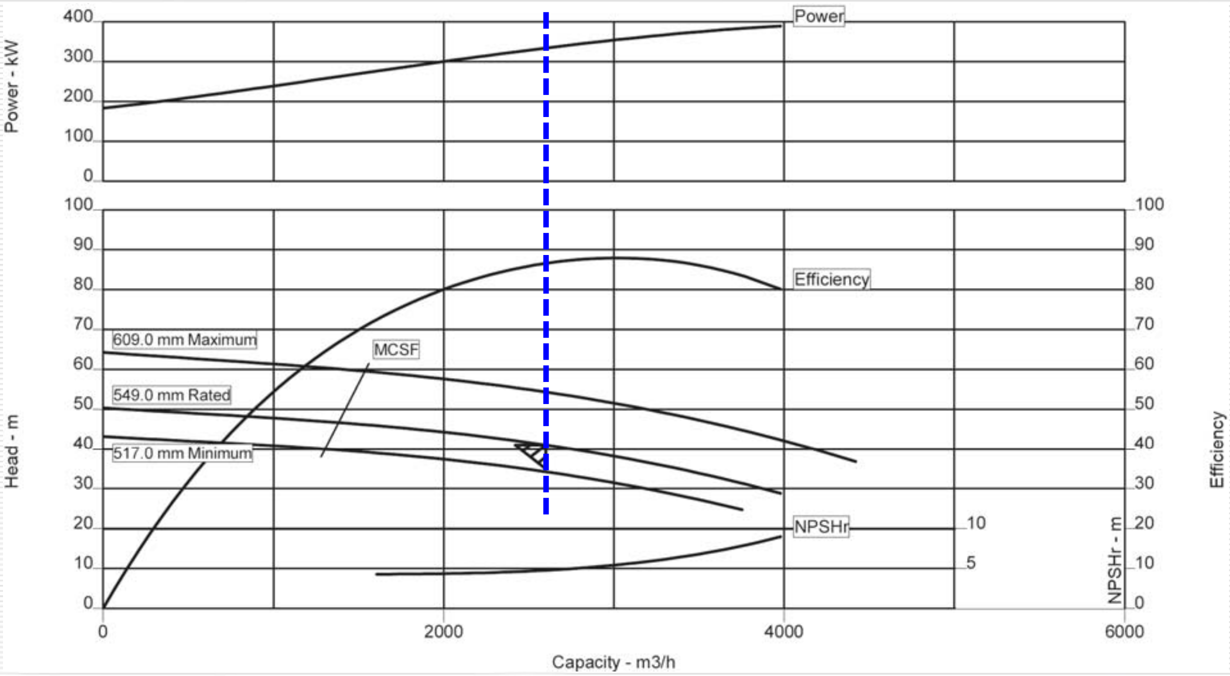
降轉速泵浦的操作模式:



$$\text{Pump 功率} = \frac{(Q) \times (H) \times (Sp.Gr.)}{(Ph) \times (Eff.)}$$

4.4泵浦的節能

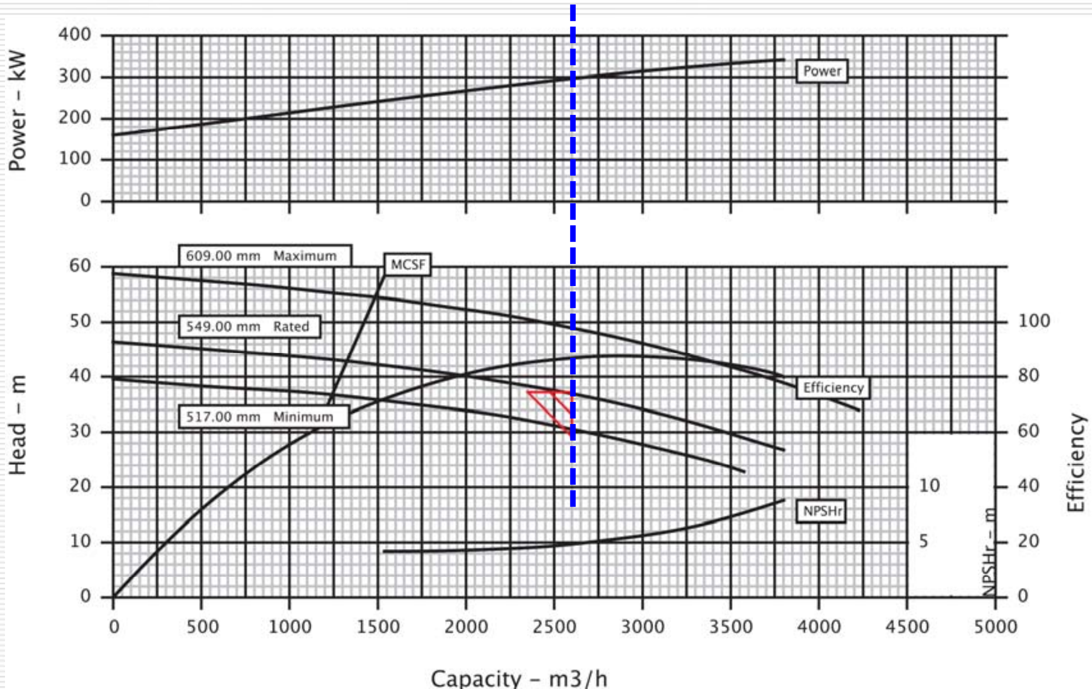
實際範例:(既有泵浦Head=41M, Power=334 Kw , 990 RPM)



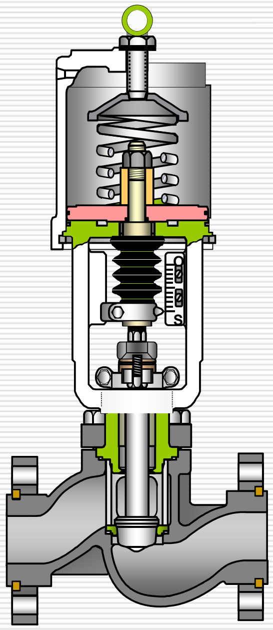
4.4泵浦的節能

實際範例:(修改泵浦Head=37M, Power=300 Kw , 948 RPM)

電力由334 Kw→300 Kw,每年節省電費:68萬



5.控制閥壓差計算



5.控制閥壓差計算

控制閥壓降之決定

$P_d = P_e + F + DP_{cv}$

Pd：泵浦出口壓力

Pe : 末端的壓力 (目的端的壓力+Static Head)

F : 管線壓降，含過濾器、熱交換器、靜態混合器，或其他元件等

- 控制閥壓降是最後決定。→ $DP_{cv} = P_d - P_e - F$
- 控制閥是壓降之Buffer Bank。
- 當流量增加時，F會升高 → DP_{cv}降低 (控制閥開度增大)

→ $P_d - P_e$ 幾乎維持不變。

- 設計時需提供控制閥足夠之壓降(一般為>0.7 kg/cm²)，以應付因為流量增加所需之壓降，否則控制閥將失去控制能力。
- Design Pump Head = Differential Head x 1.1

方法: Realistic Control-Valve Pressure Drop Chem. Eng., 94 No. 13, 1987, p.123

5.控制閥壓差計算

控制閥需預留多少壓降空間？

1) Allowance for increase flow rate → 10% → 1.1 $[(Q_{max}/Q_{design})^2-1]$ x F

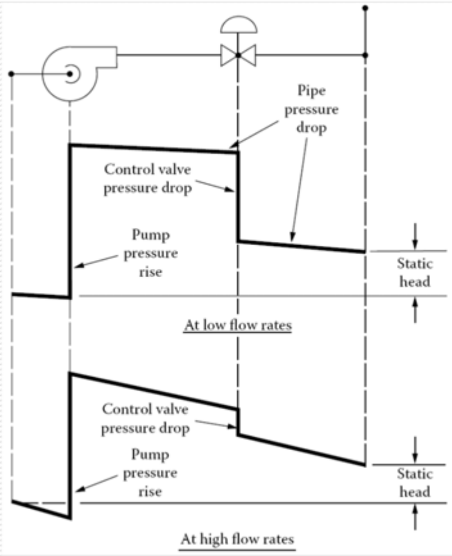
2) Allowance for possible fall off in overall system pressure drop

→ 5% of Pd

3) Allowance for control valve itself (B)：

控制閥型式	kg/cm²
Single Plug	0.77
Double Plug	0.49
Cage (Unbalanced)	0.28
Cage (Balanced)	0.28
Butterfly	0.014
V-Ball	0.07

⇨Required $DP_{cv} = 0.05Pd + 1.1 [(Q_m/Q_d)^2-1] \times F + B$



5.控制閥壓差計算

$DP_{cv} = P_d - P_e - F$ v.s. $DP_{cv} = 0.05Pd + 1.1 [(Q_m/Q_d)^2-1] \times F + B$

⇨ P_d (Pump Discharge Pressure) = $(1.1 [(Q_m/Q_d)^2-0.1] \times F + P_e + B) / 0.95$

$P_d = \{ [1.1 \times (22,000/20,000)^2-0.1] \times (0.278+0.16+0.25) + (1.41+0.01) + 0.77 \} / 0.95$

$P_d = 3.20$ kg/cm²g

泵浦入口設備端壓力　泵浦入口端之管線與元件壓損

$P_{head} = (P_d - P_s) \times 10000 / density = (3.20-(0.1+0.22-(0.035+0.032)) \times 10000/990 = 29.8$ m

泵浦入口端之Static Head

Design Pump Head = Differential Head x 1.1

Design Pump Head = 29.8 m x 1.1 = 32.8 m

Required $DP_{cv} = 0.05Ps + 1.1 [(Q_m/Q_d)^2-1] \times F + B$

$DP_{cv} = 0.05 \times 3.20 + 1.1 [(22,000/20,000)^2-1] \times (0.278+0.16+0.25) + 0.77$

$DP_{cv} = 1.09$ kg/cm²g

5.控制閥壓差計算

控制閥差壓的準則:

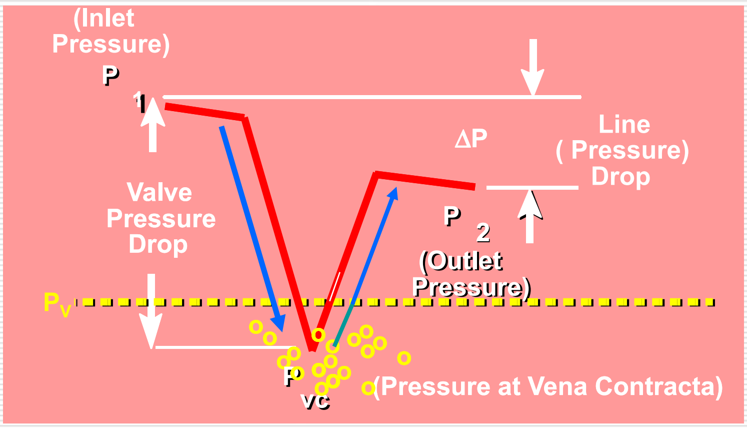
在"泵浦輸送"管線的控制閥，泵浦的揚程應該要確保控制閥能控制流量則須符合下列準則

1. 控制閥差壓在Normal流量時不建議<1.75kg/cm^2
2. 控制閥差壓至少要Normal流量下的50%總壓損
3. 控制閥差壓至少要Normal流量下10%的泵浦ΔP
   (對高ΔP(70 kg/cm^2)的泵浦，可採用7%)
4. 控制閥差壓在Rated流量時不可<0.7kg/cm^2
5. 對於泵浦供給兩個路徑以上，必須各別分開計算

5.控制閥計算(Cavitation Control)

什麼是Cavitation:

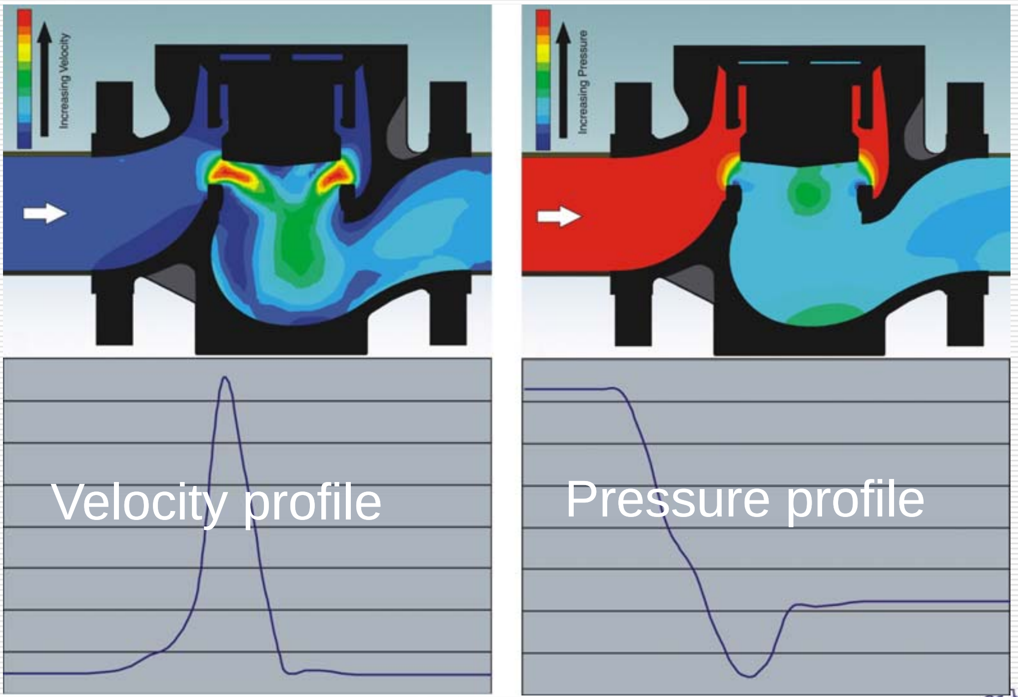
液體進入控制閥的Trim時，流速增加同時壓力下降，當液體壓力降至低於蒸氣壓時，液體開始氣化產生氣泡，當下游側壓力要回復至高於蒸氣壓時而氣泡開始被壓縮，而在Trim內壁崩潰，其產生的能量會衝擊內壁，會造成極大的Trim損壞，相對的也會產生噪音。



Single Seated Valve Pressure Recovery Diagram

5.控制閥計算(Cavitation Control)

何謂Cavitation:



Cavitation結果

- ✓Pitting and erosion
- ✓Noise and vibration
- ✓Corrosion
- ✓Combination

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5.控制閥計算(Cavitation Control)

Cavitation Damage

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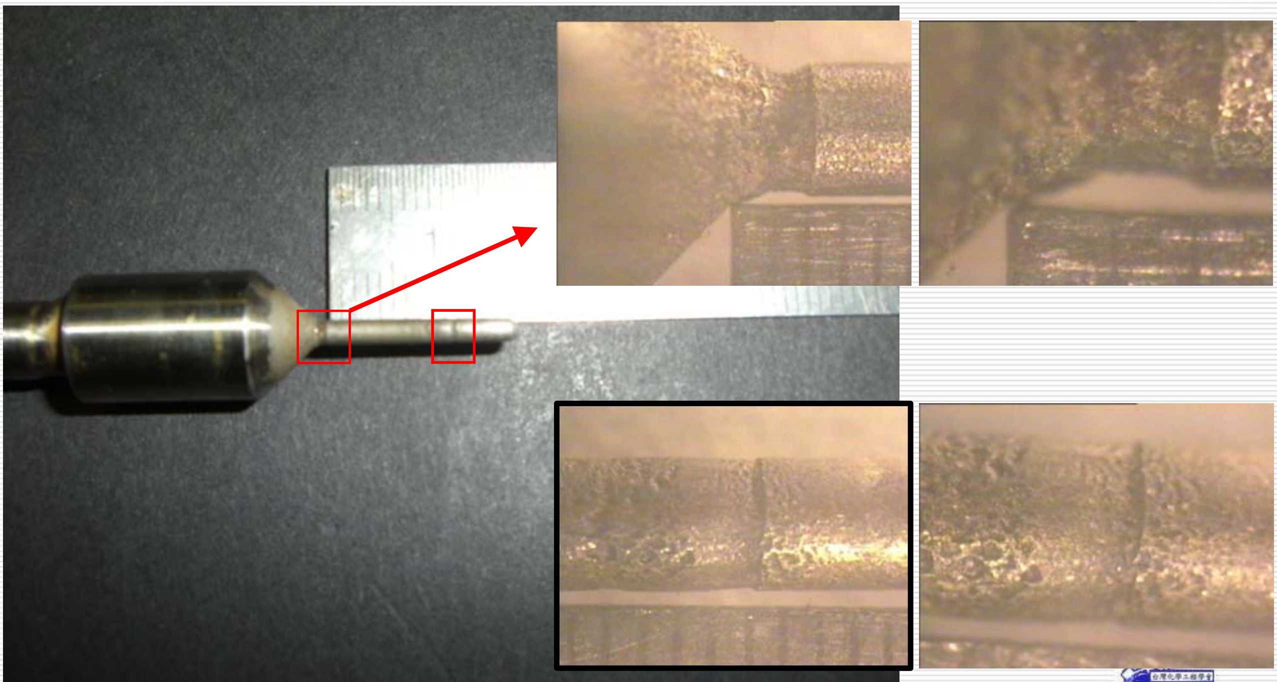
5.控制閥計算(Cavitation Control)

Cavitation Damage



5.控制閥計算(Cavitation Control)

透過實驗和實際測試結果，可使用計算Cavitaion Index (Sigma)的方式，了解Cavitation的情況並因應以消減Cavitation的發生

Cavitation Index, σ, is applied to quantify cavitation

$$\sigma = \frac{(P_1 - P_{Vap})}{(P_1 - P_2)}$$

$(P_1 - P_{Vap})$ ← Potential for resisting cavity formation

$(P_1 - P_2)$ ← Potential for causing cavity formation

P_1:控制閥上游側壓力(2直管部點), kg/cm²(a)

P_2:控制閥下游側壓力(6直管部點), kg/cm²(a)

Pvap:液體於操作溫度下的蒸氣壓, kg/cm²(a)

from: Valtek Severe Service Trims and Downstream Equipment

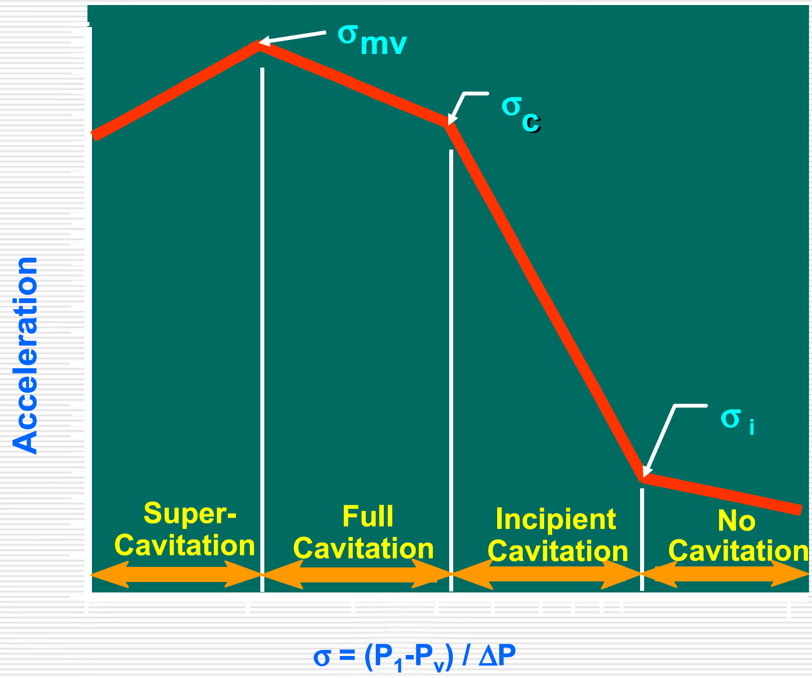
5.控制閥計算(Cavitation Control)

Cavitation Index, σ, is applied to quantify cavitation

No Cavitation σ≧2.0	No cavitation is occuring.
Incipient Cavitation 1.7 ≦ σ_i < 2.0	No cavitation control is required.(輕微Cavity形成，一般不會傷到閥)→Hardened trim provides protection!
Full Cavitation 1.5 ≦ σ_c < 1.7	Some cavitation control required. (破壞性的Cavitaion開始在此區域形成，噪音與振動將達最大值) →Mutual impingement trim may work!
Super Cavitation 1.0 ≦ σ_{mv} < 1.5	Potential for severe cavitation. (隨著Cavitaion的頻率與嚴重性增加，噪音與振動將降低) →Use staged pressure drop trim!
Flash σ≦1.0	Flashing is occuring.

5.控制閥計算(Cavitation Control)

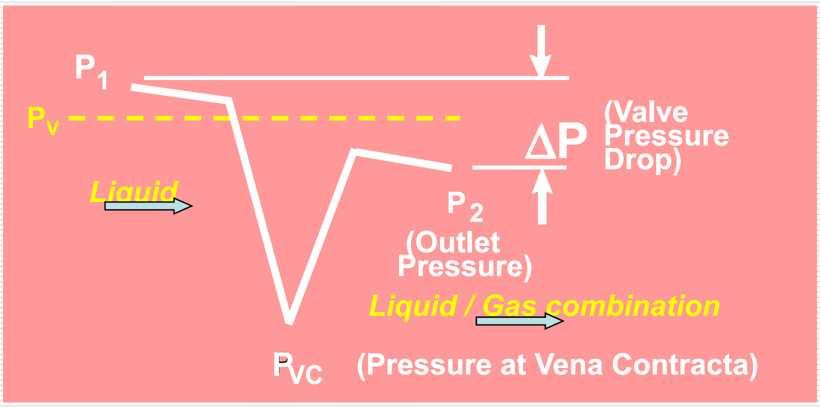
Cavitation Index, σ, is applied to quantify cavitation



5.控制閥計算(Cavitation Control)

何謂Flashing :

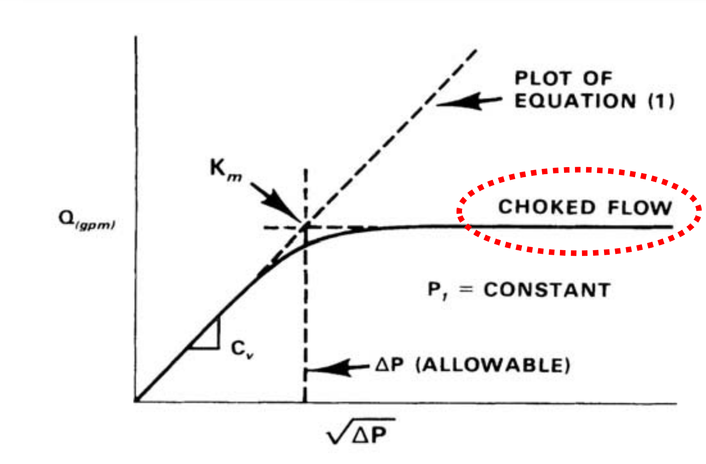
當流體流經控制閥，其壓力在蒸氣壓(Pv)以下，而在下游側其壓力未上昇到Pv時，在出口側會形成Flasing的狀態。



5.控制閥計算(Cavitation Control)

何謂Choked Flow

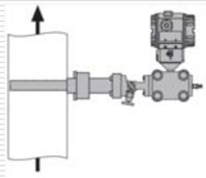
當流體流經閥體後，由於壓降而部分液體Flash產生氣泡造成流體比容積增加，而使流速增加，此時即使再增加壓降(DP)，也無法使流經閥體之流量(Q)隨之增加，形成Choked Flow 。



6.流量計壓損

6.流量計壓損

Flow Meter Type(泵浦出口常用流量計的總類):

流量計型式	流孔板		皮托管
	Orifice & DP Transmitter	Integral Orifice	Pitot Tube
圖　例			
量測性質	差壓	差壓	差壓
最低雷諾數	5000	5000	10000
準確度 (%)	±0.75%	±0.75%	±1%
流量範圍	4 / 1	4 / 1	4 / 1
壓力損失	大	大	極小
Flowrate (Water)= 20,000 kg/hr	1,600 mmH_2O	1,600 mmH_2O	0.007 kg/cm^2

依本次課程P-259出口流量(20,000 kg/hr)估算Flow Meter Pressure Drop

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6.流量計壓損

Flow Meter Type(泵浦出口常用流量計的總類):

流量計型式	渦流流量計	電磁流量計	科氏式流量計
	Vortex Meter	Magnetic Flowmeter	Coriolis Type Mass Flowmeter
圖　例			
量測性質	頻率	感應電壓	頻率差
最低雷諾數	10000	---	---
準確度 (%)	±0.75%(液體)	±0.5%	±0.15%
流量範圍	20/1 - 100/1	10/1	20/1 - 100/1
壓力損失	小	極小	大
Flowrate (Water)= 20,000 kg/hr	2", 0.078 kg/cm^2 1-1/2" ,0.227 kg/cm^2		2", 0.131 kg/cm^2 1-1/2" ,0.657 kg/cm^2

依本次課程P-259出口流量(20,000 kg/hr)估算Flow Meter Pressure Drop

6.流量計壓損

Flow Meter Pressure Drop(流量計壓損計算):

1) Orifice Flow Meter:

Orifice Flow Meter壓損必須和儀錶規格一致，若儀錶無特別指定一般則泵浦出口會採1,600 mmH_2O，流量計的流量範圍為0~2,500 mmH2O，Normal Flow於流量範圍的80%。

上述不包含液體重力流與氣體之Orifice Flow meter Pressure Drop，液體重力流與氣體是必須依製程可容許的壓損估算，或採用其他較低壓損的流量計。

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6.流量計壓損

Flow Meter Pressure Drop(流量計壓損計算):

2) Other Type:

Vortex, Coriolis (Mass), Pitot Tube, Magnetic, Turbine, Oval Gear, Thermal Mass, Ultrasonic Type等。

建議透過廠商的線上系統可計算出或請廠商Sizing，

例如(如下)

Endress+Hauser

http://www.endress.com/eh/home.nsf/#products/id/B214C864872CDE1FC1256FA4004963A

Application Sizing Flow

6.流量計壓損

Flow Meter Pressure Drop觀念:

1)流量計廠商Sizing結果：正常的情況下會小1~2個管線尺寸

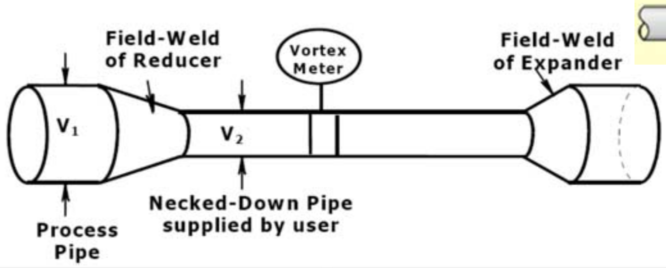
例如:管線3”，流量計可能2” or 1-1/2”

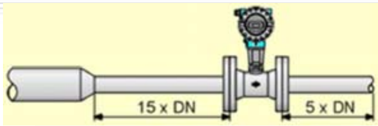
2)Straight Run(直管部):

為求流量的準確性，部分的流量計會需要直管部，所以在計算管線壓損時，必須一併考慮。

例如:Vortex, Magnetic, Turbine, Thermal Mass Type等。

Vortex Installation: Common Practice





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Q & A